PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Design and Development of a Web-Based Lot Tracking System for Golden Dragon Estate Company

In partial fulfillment of the requirements in Capstone 1 and Research

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CHAPTER 1

INTRODUCTION

1.1 Background Of The Study

The real estate industry plays a significant role in economic development by providing residential, commercial, and industrial properties to individuals and organizations. As the demand for real estate continues to grow, companies are required to manage property information efficiently to ensure accurate transactions and effective customer service. One important aspect of real estate management is monitoring the availability and status of residential lots. Accurate information regarding lot availability enables companies to respond promptly to customer inquiries, prevent double-selling, and maintain organized records of property transactions.

Golden Dragon Estate Corporation is a real estate company operating in Iloilo Province, offering residential lots in various locations such as Oton, Guimbal, and neighboring municipalities. The company currently relies on traditional methods for monitoring lot availability. Printed subdivision layout maps are used to represent residential lots, and employees manually indicate lot status through color-coding. Green represents available lots, yellow represents pending reservations, and red represents sold lots.

Although this method has been used for several years, it presents several operational challenges. Updating lot statuses requires employees to manually modify physical documents, which can be time-consuming and inefficient. As the number of lots and subdivision projects increases, maintaining accurate and updated records becomes more difficult. Furthermore, physical documents are vulnerable to damage, loss, misplacement, and deterioration over time. These issues may result in inaccurate information, delays in customer service, and potential business inefficiencies.

Advancements in information technology have enabled organizations to automate manual processes through web-based information systems. Such systems improve efficiency, accuracy, accessibility, and data security while reducing dependence on paper-based records. By digitizing lot monitoring operations, real estate companies can maintain real-time information regarding property availability and streamline internal processes.

To address the challenges encountered by Golden Dragon Estate Corporation, this study proposes the development of a Web-Based Lot Availability Monitoring System. The system will allow authorized employees to securely manage and update lot statuses through an interactive digital platform. Instead of manually coloring printed layout maps, users can simply select a lot and update its status using the system interface. The updated information will then be reflected immediately within the digital subdivision map.

The proposed system will be developed using React, Tailwind CSS, Node.js, Express.js, and MySQL. Additionally, Nodemailer SMTP integration will be utilized for email notification functionalities. Through the implementation of this system, Golden Dragon Estate Corporation can improve operational efficiency, reduce the risks associated with physical records, and enhance the accuracy of lot monitoring activities

1.2 Statement of the Problem

Golden Dragon Estate Corporation is engaged in the sale and management of residential lots across different locations within Iloilo Province, including municipalities such as Oton and Guimbal. As part of its daily operations, the company continuously monitors the availability and status of residential lots to ensure accurate information is provided to employees, management, and prospective buyers. Maintaining updated records of available, pending, and sold lots is essential for effective decision-making and efficient business operations.

At present, the company utilizes a traditional and manual method of monitoring lot availability. Printed subdivision layout maps are used as the primary reference for tracking lot statuses. Employees manually indicate the status of each lot by applying color codes on the physical maps. Green is used to represent available lots, yellow indicates pending transactions or reservations, and red signifies sold lots. While this method has been functional for many years, it presents several limitations that affect the company's operational efficiency.

One of the major concerns associated with the current system is the time and effort required to update records. Every transaction involving a lot requires employees to manually revise the printed maps, increasing the possibility of delays and inconsistencies in information. Furthermore, physical documents are vulnerable to loss, damage, deterioration, and unauthorized alterations. As the company continues to manage multiple subdivision projects and an increasing number of lot records, the challenges associated with manual monitoring become more evident.

The absence of a centralized digital system also limits the accessibility and reliability of information. Employees must rely on physical records when verifying lot availability, which can make the retrieval of information slower and less efficient. Additionally, there is a greater possibility of human error in recording, updating, and maintaining lot information. These issues may negatively affect customer service, internal operations, and overall productivity.

In response to these challenges, the researchers propose the development of a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. The proposed system aims to automate the monitoring and management of lot information through a secure and centralized platform. By replacing the manual process with a digital solution, the company can improve efficiency, enhance data accuracy, strengthen security, and facilitate easier access to information.

General Problem

How can a Web-Based Lot Availability Monitoring System be developed and implemented to improve the efficiency, accuracy, security, accessibility, and reliability of monitoring residential lot availability at Golden Dragon Estate Corporation?

Specific Problems

Specifically, this study seeks to answer the following questions:

1. The current lot monitoring process relies heavily on printed subdivision maps and manual color-coding methods.
2. Updating lot statuses requires considerable time and effort from employees, resulting in operational inefficiencies.
3. Physical records are susceptible to loss, damage, deterioration, and misplacement.
4. The existing process increases the likelihood of human errors during the recording and updating of lot information.
5. The company lacks a centralized database for organizing and managing lot records.
6. Access to lot information is limited to physical documents, making information retrieval less convenient and efficient.
7. Monitoring lot availability across multiple subdivision projects becomes increasingly difficult as records grow.
8. The current process does not provide real-time updates regarding changes in lot status.

9. There is no secure authentication mechanism that restricts access to authorized personnel only.
10. The company lacks an automated system capable of maintaining accurate and organized lot information.
11. The existing method makes it difficult to maintain consistency and reliability in monitoring available, pending, and sold lots.
12. The current process does not utilize modern information technology solutions that could improve operational efficiency.

1.3 Objectives of the Study

The primary purpose of this study is to develop a modern technological solution that addresses the challenges encountered by Golden Dragon Estate Corporation in monitoring residential lot availability. Through the implementation of a web-based information system, the researchers aim to improve the company's operational processes by automating manual tasks and providing a more efficient method of managing lot records.

The proposed system seeks to transform traditional paper-based monitoring procedures into a centralized digital platform capable of providing accurate, reliable, and real-time information regarding lot availability. By integrating database management, authentication mechanisms, and interactive monitoring features, the system is expected to improve overall organizational productivity and data management practices.

1.3.1 General Objective

The general objective of this study is to design, develop, and implement a Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation that will automate the monitoring, management, and updating of residential lot information while improving operational efficiency, data accuracy, accessibility, and security.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Design and develop a web-based lot availability monitoring system for Golden Dragon Estate Corporation.
2. Create a centralized database for storing, managing, and retrieving lot information efficiently.
3. Develop an interactive digital subdivision layout that visually represents lot availability.
4. Automate the process of updating lot statuses from available to pending or sold.
5. Implement secure user authentication and authorization features for authorized employees.
6. Allow users to efficiently search, filter, and retrieve lot information.
7. Provide real-time updates whenever changes are made to lot records.
8. Reduce the organization's dependence on paper-based records and manual monitoring procedures.

9. Improve the accuracy and consistency of lot information management.
10. Minimize the occurrence of human errors associated with manual record keeping.
11. Enhance data security through controlled system access.
12. Improve employee productivity by simplifying lot monitoring and management activities.
13. Integrate email notification functionality using Nodemailer SMTP for relevant system notifications.
14. Generate organized and accessible records that can support management decision-making.
15. Provide a user-friendly interface that is easy to learn and use by authorized personnel.

1.4 Scope and Delimitation

This study focuses on the development of a Web-Based Lot Availability Monitoring System specifically designed for Golden Dragon Estate Corporation. The proposed system will serve as a centralized platform for monitoring, managing, and updating residential lot information across the company's subdivision projects.

The system will provide functionalities that support the daily monitoring activities of authorized employees and administrators. Through the use of modern web technologies such as React, Tailwind CSS, Node.js, Express.js, and MySQL, the system will provide a secure and efficient environment for managing lot records.

Specifically, the system will include the following features:

- User login and authentication
- Role-based access control
- User account management
- Lot information management
- Interactive subdivision layout visualization
- Lot status monitoring
- Status updates for available, pending, and sold lots
- Search and filtering capabilities
- Database storage and management
- Email notification integration through Nodemailer SMTP
- Administrative monitoring functions
- Secure record management

The proposed system will be accessible through web browsers and will be deployed as a web-based application.

Delimitation of the Study

Despite its intended benefits, the proposed system has several limitations.

1. The system focuses exclusively on monitoring residential lot availability.
2. The system is intended only for internal organizational use.
3. Access will be limited to authorized employees, administrators, and management personnel.
4. Clients and prospective buyers will not have direct access to the system.
5. The system will not support online reservations or online purchasing of lots.
6. Payment processing and financial transactions are excluded from the system.
7. Accounting and bookkeeping functionalities are beyond the scope of the study.
8. Legal document management and contract processing are not included.
9. Mobile application development is not part of the project.
10. GIS-based mapping and satellite visualization technologies will not be implemented.
11. The system will only cover subdivision projects managed by Golden Dragon Estate Corporation.
12. Predictive analytics, artificial intelligence, and sales forecasting functionalities are not included.

Target Users

The intended users of the proposed system include:

- System Administrators
- Authorized Employees
- Management Personnel

Platform

The proposed system will be developed as a web-based application accessible through desktop and laptop web browsers connected to the internet or a local network.

1.5 Significance of the Study

The successful development and implementation of the proposed Web-Based Lot Availability Monitoring System is expected to provide significant benefits to Golden Dragon Estate Corporation and its stakeholders. The system aims to improve operational efficiency, enhance data accuracy, and reduce the challenges associated with manual record management.

Golden Dragon Estate Corporation

The organization will benefit from a more efficient and reliable method of monitoring residential lot availability. The proposed system will improve record management, reduce operational delays, and support the company's efforts to modernize its business processes.

Administrators

Administrators will benefit from having centralized control over system operations, user accounts, and lot records. The system will enable them to efficiently manage information while maintaining security and accuracy.

Employees

Authorized employees will experience a more convenient method of updating and monitoring lot statuses. The automation of manual processes will reduce workload, save time, and improve productivity.

Management Personnel

Management personnel will gain access to organized and updated information regarding lot availability. This will support decision-making, monitoring activities, and strategic planning efforts.

Clients and Prospective Buyers

Although clients will not directly access the system, they will indirectly benefit from faster and more accurate information regarding lot availability, resulting in improved customer service.

Researchers

The study will provide researchers with valuable knowledge and practical experience in software development, web technologies, database management, systems analysis, and project implementation.

Future Researchers

Future researchers may utilize this study as a reference for developing similar systems related to property management, monitoring systems, and web-based information systems.

Future Developers

Future developers may use the system design, methodologies, and findings of this study as a foundation for creating enhanced versions of the system with additional features and capabilities.

1.6 Definition of Terms

Administrator

A user who possesses the highest level of access privileges and is responsible for managing system operations, user accounts, and records.

Authentication

The process of verifying the identity of a user before granting access to the system.

Authorization

The process of determining the access rights and permissions granted to a user after successful authentication.

Centralized Database

A database that stores information in a single location where data can be managed, accessed, and updated efficiently.

Database

An organized collection of data stored electronically for easy access, retrieval, and management.

Express js

A web application framework for Node.js used to build server-side functionalities and application programming interfaces (APIs).

Lot Availability

The current status of a residential lot indicating whether it is available, pending, or sold.

Monitoring System

A computerized application designed to track, manage, and display information for monitoring purposes.

Node js

A JavaScript runtime environment used to develop scalable server-side applications.

Nodemailer SMTP

A Node.js module that enables applications to send emails through the Simple Mail Transfer Protocol (SMTP).

React

A JavaScript library used for building interactive and dynamic user interfaces.

Residential Lot

A parcel of land intended for residential development and offered for sale by a real estate company.

Tailwind CSS

A utility-first CSS framework used for designing responsive and visually appealing web interfaces.

User Authentication

A security mechanism that validates user credentials before allowing access to the system.

Web-Based Application

A software application that operates through a web browser and can be accessed via a network connection.

CHAPTER 2

2.1 Foreign Studies

The advancement of information technology has significantly influenced the real estate industry worldwide. Researchers have developed various web-based and database-driven systems to improve property management, information accessibility, and operational efficiency. These studies provide valuable insights into the effectiveness of digital technologies in managing real estate information and serve as a foundation for the development of the proposed Web-Based Lot Availability Monitoring System.

#1 Amit Kumar (2024) Development and Implementation of a Real Estate Web Application for Modern Web Application

The real estate industry has experienced significant technological advancements in recent years, driven by the increasing demand for efficient property management and digital accessibility. Recognizing these industry developments, Kumar (2024) developed and implemented a Real Estate Management System (REMS) designed to modernize the management of property-related information through a web-based platform.

The study emphasized the importance of integrating digital technologies into real estate operations to address inefficiencies associated with traditional methods of record-keeping and property monitoring.

According to Kumar (2024), conventional real estate management practices often rely on manual documentation, fragmented information storage, and localized record management systems. These approaches frequently result in delayed information updates, data inconsistencies, communication gaps among stakeholders, and difficulties in maintaining accurate property records. To address these concerns, the researcher proposed a comprehensive web application that centralizes property information within a unified database environment. Through this centralized architecture, users can access, update, and manage property records in real time, thereby improving operational efficiency and decision-making processes.

The developed Real Estate Management System incorporated several automated functionalities, including property listing management, real-time property status updates, user account management, and information retrieval features. The system was designed to streamline workflows by reducing manual intervention and providing stakeholders with immediate access to accurate property data. The study further highlighted that automated listing modules enable organizations to maintain updated information regarding property descriptions, pricing details, availability status, and other essential records without relying on traditional paper-based processes.

One of the significant contributions of Kumar's study is its demonstration of how centralized databases and web-based technologies can improve data accuracy and accessibility within the real estate sector. The implementation of the system showed that digital platforms can effectively reduce administrative workload, minimize human errors, and improve communication among users involved in property management activities. Furthermore, the study emphasized the role of modern web applications in supporting organizational growth through scalable and maintainable information management solutions.

The findings of the study indicate that organizations adopting web-based real estate management systems can achieve greater operational efficiency, enhanced record management capabilities, and improved service delivery. By providing a centralized repository for property-related information, the system ensures that data remains consistent, accessible, and secure. These advantages contribute to more effective property monitoring and facilitate faster responses to changing business requirements.

Relevance to the Current Study

The study conducted by Kumar (2024) is highly relevant to the present research because both studies focus on the application of web-based technologies to improve real estate management processes. Similar to the Real Estate Management System developed by Kumar, the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation seeks to replace manual procedures with a centralized digital platform that supports efficient information management and real-time data updates.

The current study specifically addresses the challenges encountered by Golden Dragon Estate Corporation in monitoring residential lot availability through manually maintained and color-coded subdivision maps. Kumar's findings provide strong evidence that centralized databases and automated management systems can significantly improve data accessibility, accuracy, and operational efficiency. These concepts directly support the development of the proposed system, which aims to allow authorized personnel to update lot statuses electronically and monitor lot availability through an interactive web interface.

While Kumar's research focused on broader real estate management functionalities, including property listings and transaction-related information, the current study concentrates specifically on lot availability monitoring and status management. Despite this difference in scope, both studies share the common objective of utilizing digital solutions to enhance real estate operations. Therefore, Kumar's work serves as an important foundation that supports the need for digital transformation within real estate organizations and validates the adoption of web-based monitoring systems as effective tools for modern property management.

#2 **Kushal B. C. (2026) EstateManager: A JWT-Secured Real Estate Platform with Role-Based Access Control**

As the real estate industry continues to embrace digital transformation, ensuring the security and integrity of property-related information has become a significant concern. The transition from paper-based documentation to web-based information systems introduces new challenges related to unauthorized access, data manipulation, and information security. Addressing these concerns, Kushal B. C. (2026) developed EstateManager, a secure web-based real estate platform designed to provide efficient property management while implementing advanced security mechanisms to protect sensitive data.

The study emphasized that many small and medium-sized real estate organizations continue to rely on fragmented workflows and manual record-keeping procedures. Such practices often result in inconsistent data management, limited accountability, and increased vulnerability to human error. Furthermore, the absence of proper access control mechanisms may expose property records to unauthorized modifications, potentially affecting the reliability and credibility of organizational data. Recognizing these challenges, the researcher proposed a centralized digital platform that combines efficient property management functions with robust security features.

EstateManager was developed using a full-stack architecture integrated with a MySQL database management system. One of the key features of the platform is the implementation of JSON Web Token (JWT) authentication, which ensures that only verified users can access protected resources within the application. JWT provides a secure method of transmitting authentication information between users and the server, reducing the risk of unauthorized access. In addition, the platform incorporates Role-Based Access Control (RBAC), allowing administrators to assign specific permissions based on user roles and responsibilities.

The implementation of RBAC enables the system to regulate access to critical operations such as property creation, editing, updating, and deletion. Through this mechanism, users are restricted to functions that correspond to their designated authority levels. This approach not only strengthens system security but also improves accountability by ensuring that all modificationscan be traced to authorized personnel. The study further demonstrated that the integration of authentication and authorization mechanisms contributes to the protection of sensitive real estate records while maintaining system usability.

Another significant contribution of the study is its focus on performance optimization. According to the findings, the system achieved efficient data retrieval and filtering capabilities, maintaining an average query response time of less than 100 milliseconds. This level of performance indicates that secure systems can effectively balance data protection and operational efficiency without negatively affecting user experience. The researcher concluded that implementing robust security measures does not necessarily compromise system responsiveness, making secure web applications practical for real-world real estate operations.

The results of the study demonstrated that EstateManager successfully improved the security, accessibility, and management of property records. By integrating centralized database management, secure authentication, role-based permissions, and efficient system performance, the platform provided a reliable solution for modern real estate organizations seeking to digitize their operations while safeguarding critical information.

Relevance to the Current Study

The study conducted by Kushal B. C. (2026) is highly relevant to the present research because both studies involve the development of web-based systems designed to manage and monitor real estate information. Similar to EstateManager, the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation aims to centralize real estate data and provide authorized users with efficient access to property-related information through a secure digital platform.

One of the most significant contributions of Kushal's study to the current research is its emphasis on authentication and access control mechanisms. Since the proposed system allows users to update critical lot statuses, including changing lots from available to pending or sold, implementing secure authentication procedures is essential. The use of Role-Based Access Control provides a valuable framework for regulating user permissions and ensuring that only authorized personnel can modify lot records. This helps protect the integrity and accuracy of the company's real estate inventory.

Additionally, the study supports the importance of maintaining a secure database environment where sensitive business information is protected against unauthorized access and accidental modifications. The findings reinforce the need for strong security practices within web-based monitoring systems, particularly in industries that rely heavily on accurate and up-to-date records. The efficient performance demonstrated by EstateManager also highlights the importance of designing systems that can process information quickly while maintaining security standards.

Although EstateManager focuses on broader property management functions, whereas the current study concentrates specifically on lot availability monitoring, both systems share common objectives related to data management, system security, and operational efficiency. Therefore, Kushal's research serves as an important reference that supports the implementation of secure user authentication, role-based access control, and centralized database management within the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation.

#3 Joshi, K., Sidhu, K. S., Malik, N., Jadhav, A., Saini, D. K. J. B., et al. (2024) Innovative Real Estate Management System: Artificial Intelligence-Based Segregation

The rapid advancement of information technology has significantly transformed the real estate sector by introducing intelligent systems capable of automating property management processes and enhancing decision-making capabilities. In response to the growing complexity of managing large volumes of real estate data, Joshi et al. (2024) developed an Innovative Real Estate Management System that integrates modern web technologies and artificial intelligence techniques to improve the organization, accessibility, and management of property information. The study focused on developing a scalable and responsive platform capable of supporting real-time operations within the real estate industry.

According to the researchers, traditional real estate management methods often struggle to accommodate the increasing volume of property records, user interactions, and transactional activities generated in modern business environments. As organizations expand their property portfolios, manual management approaches become increasingly inefficient and difficult to maintain. To address these challenges, the study proposed a web-based system built on a JavaScript-centered technology stack consisting of React, Node.js, and Express.js.

This architecture was selected because of its flexibility, scalability, and ability to support real-time data processing. One of the primary contributions of the study is its demonstration of how Node.js can effectively handle asynchronous and event-driven operations within a real estate environment. By utilizing non-blocking input and output processes, the system was capable of managing multiple requests simultaneously without significant reductions in performance. This characteristic is particularly important in applications that require frequent updates, continuous user interaction, and rapid retrieval of information. The researchers emphasized that responsive server-side processing contributes significantly to overall system efficiency and user satisfaction.

In addition to its scalable architecture, the system incorporated comprehensive user authentication mechanisms to protect sensitive real estate information. Authentication procedures were implemented to ensure that only authorized individuals could access specific functions and records within the application. This security framework enhanced data integrity and reduced the risk of unauthorized modifications. The study also integrated data visualization tools and interactive dashboards that allowed administrators to monitor property information through graphical representations and summarized reports.

The interactive dashboard component was identified as one of the most valuable features of the system. Through the use of visual analytics, administrators could quickly interpret large amounts of property-related information and identify important trends, patterns, and operational indicators. The researchers found that dashboard-based reporting significantly improved decision-making processes by presenting complex information in a clear and organized manner.

This capability enabled administrators to manage resources more effectively while reducing the time required to analyze operational data.

The findings of the study demonstrated that modern web technologies can successfully support real-time real estate management applications. The combination of React, Node.js, and Express.js provided a reliable foundation for building scalable systems capable of handling continuous data transactions and user interactions. Furthermore, the incorporation of secure authentication mechanisms and dashboard reporting tools contributed to enhanced operational efficiency, improved data accessibility, and better overall system performance.

The researchers concluded that adopting modern web-based architectures allows real estate organizations to improve information management, strengthen security measures, and support future organizational growth. The study highlights the importance of scalable and responsive technologies in meeting the demands of contemporary property management environments.

Relevance to the Current Study

The study conducted by Joshi et al. (2024) is highly relevant to the present research because both studies focus on the development of web-based systems designed to improve real estate management through modern information technologies. Similar to the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation, the researchers utilized a full-stack JavaScript architecture composed of React, Node.js, and Express.js to develop a responsive and scalable platform capable of managing property-related information.

A significant contribution of this study to the current research is its validation of the technology stack selected for system development. Since the proposed capstone project also utilizes React for the front-end interface and Node.js with Express.js for back-end operations, the findings of Joshi et al. provide strong technical support for the chosen development framework. The successful implementation of these technologies demonstrates their capability to handle real-time updates, user interactions, and database transactions efficiently.

The study also reinforces the importance of interactive dashboards in supporting administrative decision-making. Similar to the dashboard functionalities proposed for the lot availability monitoring system, the dashboard developed by Joshi et al. provided summarized information and visual representations that allowed administrators to quickly understand operational data. This concept is directly applicable to the current project, where authorized personnel must monitor available, pending, and sold lots efficiently.

Furthermore, the emphasis on secure authentication mechanisms aligns closely with the security requirements of Golden Dragon Estate Corporation. Since the proposed system involves updating critical lot status information, implementing secure access controls is essential to prevent unauthorized modifications and maintain data integrity.

Although the study explored broader real estate management functions and incorporated artificial intelligence-based segregation techniques, while the current study focuses specifically on lot availability monitoring, both projects share the common objective of improving operational efficiency through web-based technologies. Consequently, the research of Joshi et al. (2024) serves as a valuable reference that supports the technical architecture, security framework, and dashboard functionalities proposed for the Web-Based Lot Availability Monitoring System.

#4 Mandar Sharad Salunkhe (2026) Property Rental & Management System

The increasing demand for efficient property management solutions has encouraged real estate organizations to adopt digital technologies that streamline operations and improve information accessibility. As businesses manage larger volumes of property-related data, traditional paper-based systems become less effective due to their limitations in data storage, retrieval, and maintenance. Recognizing these challenges, Salunkhe (2026) developed a web-based Property Rental and Management System aimed at modernizing property administration processes through the integration of contemporary web technologies and centralized database management.

The study emphasized that many real estate and rental management organizations continue to rely on manual documentation and fragmented record-keeping practices. These conventional methods often result in delays in information retrieval, duplication of records, increased administrative workload, and a greater likelihood of human error. Furthermore, physical documents are susceptible to loss, damage, and unauthorized access, creating additional risks for organizations that depend on accurate property information. To address these concerns, the researcher proposed a centralized web-based platform capable of storing, managing, and retrieving property information efficiently.

The developed system utilized JavaScript-based technologies for application development and a MySQL relational database management system for centralized data storage. The integration of Node.js on the server side enabled the application to process requests efficiently while maintaining real-time communication between users and the database. This architecture allowed users to access current property information from multiple devices and locations, thereby improving accessibility and operational flexibility. By centralizing all records within a single database environment, the system eliminated the need for maintaining multiple copies of documents and significantly reduced the possibility of data inconsistencies.

One of the key findings of the study was the positive impact of centralized data management on organizational efficiency. The implementation of the Property Rental and Management System resulted in faster access to property records, improved accuracy of information, and reduced dependency on manual administrative procedures. The researcher observed that users could retrieve property information more quickly and perform management tasks with greater efficiency compared to traditional paper-based workflows. These improvements contributed to better decision-making processes and enhanced overall productivity.

The study also highlighted the importance of real-time information availability in modern property management. Through the use of web-based technologies and database-driven operations, updates to property information became immediately accessible to authorized users. This capability ensured that records remained current and consistent, reducing the likelihood of outdated information being used in business operations. Additionally, the centralized structure facilitated better record organization and simplified long-term data maintenance.

The findings demonstrated that web-based management systems can significantly improve the effectiveness of property administration by automating routine processes, enhancing information accessibility, and strengthening data integrity. The researcher concluded that organizations adopting centralized digital platforms are better positioned to manage growing volumes of property information while maintaining operational efficiency and accuracy.

Relevance to the Current Study

The study conducted by Salunkhe (2026) is highly relevant to the present research because both studies focus on replacing traditional manual processes with centralized web-based systems designed to improve the management of real estate information. Similar to the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation, the Property Rental and Management System was developed to address the limitations of paper-based record management and improve operational efficiency through digital transformation.

One of the most significant contributions of Salunkhe's study to the current research is its validation of centralized database management using Node.js and MySQL technologies. Since the proposed lot availability monitoring system utilizes the same technological combination, the study provides strong evidence supporting the effectiveness of these tools in managing real-time property information. The successful implementation of a Node.js and MySQL architecture demonstrates its capability to support secure, scalable, and efficient information management within real estate environments.

The study also reinforces the importance of real-time accessibility and centralized record storage. Golden Dragon Estate Corporation currently relies on manually maintained subdivision maps to monitor lot availability, creating challenges related to information updates and document management. The findings of Salunkhe indicate that centralized digital systems can eliminate these inefficiencies by providing users with immediate access to updated information through a single platform. This concept directly supports the primary objective of the current study, which is to enable authorized personnel to monitor and update lot statuses electronically.

Furthermore, the reduction of manual workload observed in Salunkhe's study aligns closely with the intended benefits of the proposed system. By automating lot monitoring activities and replacing paper-based records with digital data management, the proposed capstone project seeks to improve productivity, minimize human errors, and enhance the accuracy of real estate inventory information.

Although Salunkhe's research primarily focused on property rental and management operations, while the current study concentrates specifically on lot availability monitoring, both projects share common goals related to data centralization, operational efficiency, and digital record management. Therefore, the study serves as a valuable reference that supports the technological framework and operational objectives of the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation.

#5 Vijayalakshmi, S., Yasini, V., Sreeja, G., & Sneka, A. (2026) PROPVISTA: Real Estate Website

The rapid evolution of web technologies has significantly transformed how real estate organizations manage property information and interact with users. Modern real estate platforms are expected to provide not only centralized property management but also interactive user experiences, automated notifications, and efficient communication mechanisms. Recognizing these demands, Vijayalakshmi et al. (2026) developed PROPVISTA, a web-based real estate management platform designed to enhance property exploration, information management, and administrative operations through the integration of contemporary web development technologies.

The researchers emphasized that traditional property management systems often require extensive manual monitoring and administrative intervention to maintain updated records and communicate important information. Such practices can result in delays, increased workload, and a higher probability of errors in information dissemination. To address these challenges, PROPVISTA was developed using a modern full-stack architecture composed of React for the front-end interface, Express.js for server-side routing, and Node.js for backend processing. This technology stack enabled the creation of a highly responsive and interactive web application capable of handling real-time user interactions and data updates.

One of the major features highlighted in the study is the implementation of automated data visualization dashboards. These dashboards provide users and administrators with summarized information presented through intuitive graphical components and organized data displays. The researchers found that visual representations of property information significantly improved the ability of administrators to monitor records, identify trends, and make informed decisions. By reducing the need to manually review large volumes of data, dashboard-based reporting contributed to increased efficiency and improved operational management.

Another significant contribution of the study is its integration of secure user authentication mechanisms and automated notification features. The researchers implemented authentication controls to ensure that access to system functionalities was restricted to authorized users. This security framework helped protect sensitive property information while maintaining data integrity. Additionally, automated notifications were incorporated to provide users with timely updates regarding property activities, reducing dependence on manual communication processes.

The study revealed that combining interactive user interfaces with automated notification systems substantially improved workflow efficiency. Administrators were able to receive important updates immediately, enabling them to respond more quickly to changes within the system. Furthermore, automated notifications minimized the risk of overlooked information and ensured that critical updates were communicated consistently. These features contributed to improved coordination among stakeholders and enhanced overall system reliability.

The findings demonstrated that modern real estate applications benefit significantly from the integration of responsive user interfaces, dashboard reporting, automated notifications, and secure authentication mechanisms. The researchers concluded that these technologies not only improve user experience but also strengthen organizational efficiency by reducing manual workloads and enhancing information accuracy. As a result, PROPVISTA serves as an example of how web-based solutions can support digital transformation within the real estate industry.

Relevance to the Current Study

The study conducted by Vijayalakshmi et al. (2026) is highly relevant to the present research because both studies focus on utilizing modern web technologies to improve the management and monitoring of real estate information. Similar to PROPVISTA, the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation employs a full-stack architecture that utilizes React for the front-end interface and Node.js with Express.js for backend operations. This similarity provides strong technical support for the development approach adopted in the current study.

One of the most important contributions of the study is its emphasis on dashboard reporting and automated notification systems. The proposed lot availability monitoring system also incorporates dashboard functionalities designed to provide administrators with summarized information regarding available, pending, and sold lots. Through visual representations and real-time updates, authorized personnel can efficiently monitor the status of residential lots and make informed decisions regarding inventory management.

Furthermore, the automated notification mechanisms discussed in PROPVISTA directly support the inclusion of email notification services within the proposed system. Golden Dragon Estate Corporation requires timely updates regarding changes in lot availability, and the integration of Nodemailer SMTP allows the system to automatically notify authorized users whenever significant updates occur. This reduces dependence on manual communication and ensures that stakeholders remain informed about critical system activities.

The study also reinforces the importance of secure authentication and controlled access to property records. Since the proposed system manages sensitive real estate information, implementing secure login procedures and user access controls is essential to maintain data integrity and prevent unauthorized modifications. The findings of Vijayalakshmi et al. validate the effectiveness of combining security measures with interactive web technologies to create reliable and efficient management systems.

Although PROPVISTA was designed as a broader real estate platform that supports property exploration and management, while the current study focuses specifically on lot availability monitoring, both systems share common objectives related to automation, data accessibility, user interaction, and operational efficiency. Therefore, the study serves as a valuable reference that supports the technological architecture, dashboard reporting features, notification services, and security mechanisms incorporated within the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation.

#6 D. Wagh (2024) Real Estate Website

The development of modern real estate management systems requires not only effective technological solutions but also a structured software development methodology capable of accommodating changing requirements and continuous improvements. Recognizing the dynamic nature of real estate operations, Wagh (2024) developed a web-based Real Estate Management System using the Agile Software Development framework. The study focused on creating a flexible and secure platform that allows administrators to efficiently manage property information while ensuring data integrity through controlled user access.

According to Wagh (2024), traditional property management methods often suffer from inefficiencies associated with manual record maintenance, delayed information updates, and difficulties in monitoring inventory status. As real estate organizations continue to expand, these limitations become increasingly problematic because they hinder operational efficiency and increase the likelihood of human error. To address these concerns, the researcher proposed a web-based solution that centralizes property records within a MySQL database and provides authorized users with real-time access to inventory information.

A significant aspect of the study was the adoption of the Agile Software Development Methodology. The researcher emphasized that Agile promotes iterative development, continuous testing, and regular stakeholder feedback throughout the software development lifecycle. By dividing the project into manageable development cycles, the system could be improved incrementally based on user requirements and emerging organizational needs. This approach enhanced project flexibility and increased the likelihood of delivering a system that effectively addresses operational challenges.

The developed Real Estate Management System incorporated a MySQL database to manage property records and inventory information. Through a centralized database architecture, administrators could update property availability statuses instantly, ensuring that records remained accurate and up to date. The study highlighted the importance of real-time inventory management, particularly in environments where property information frequently changes. By providing immediate access to updated records, the system improved communication among stakeholders and reduced the delays commonly associated with manual record-keeping processes.

Another major contribution of the study was its implementation of Role-Based Access Control (RBAC). The researcher identified data security as a critical concern in digital property management systems, especially when multiple users are granted access to sensitive information. To mitigate the risk of unauthorized modifications, the application assigned permissions based on user roles and responsibilities. This approach ensured that only designated administrators could modify property statuses or perform critical system functions. As a result, the integrity, accuracy, and security of property records were significantly enhanced.

The findings of the study demonstrated that combining Agile development practices with secure access control mechanisms contributes to the successful implementation of web-based real estate systems. The researcher observed improvements in operational efficiency, data accuracy, and user accountability. Furthermore, the system reduced reliance on manual processes and enabled organizations to manage property inventories more effectively through centralized and secure digital records.

The study concluded that web-based property management systems developed using Agile methodologies offer significant advantages in terms of flexibility, maintainability, and responsiveness to organizational requirements. Additionally, the incorporation of role-based security controls ensures that sensitive real estate information remains protected while still being accessible to authorized personnel.

Relevance to the Current Study

The study conducted by Wagh (2024) is highly relevant to the present research because it provides both a methodological and functional foundation for the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Similar to Wagh's system, the current study seeks to replace manual property monitoring practices with a centralized digital platform that enables real-time inventory management and secure information access.

One of the most important contributions of the study is its validation of the Agile Software Development Methodology. Since the proposed capstone project also utilizes Agile as its development framework, Wagh's research provides strong support for the chosen methodology. The iterative nature of Agile allows developers to continuously refine system features based on stakeholder feedback, ensuring that the final product aligns closely with organizational requirements. This approach is particularly beneficial for Golden Dragon Estate Corporation, where user needs may evolve during the development process.

The study also reinforces the importance of centralized database management through MySQL. Similar to the proposed lot monitoring system, Wagh's application utilized a MySQL database to store and manage inventory information. The successful implementation of this database structure demonstrates its suitability for handling real estate records, maintaining data consistency, and supporting real-time updates. These findings directly support the decision to use MySQL as the database management system for the current project.

Furthermore, the implementation of Role-Based Access Control closely aligns with the security requirements of Golden Dragon Estate Corporation. Since the proposed system allows users to update lot statuses such as available, pending, and sold, it is essential that only authorized personnel have access to these functions. Wagh's findings highlight the effectiveness of RBAC in preventing unauthorized modifications and maintaining the integrity of inventory records. This concept serves as a valuable reference for the security framework of the proposed system.

Although Wagh's research focused on a broader real estate management platform, while the current study specifically addresses lot availability monitoring, both systems share common objectives related to inventory management, real-time status updates, centralized data storage, and secure access control. Therefore, the study provides strong methodological and technical support for the development of the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation.

#7 S. Paidi (2025) Web-Based Real Estate Management System: DBMS-Supported Solution for Property Listings and Buyer-Seller Interaction

The growing complexity of real estate operations has created a demand for digital systems capable of efficiently managing large volumes of property information while ensuring data accuracy and accessibility. Traditional methods of property management often rely on manual records, paper-based documentation, and disconnected information repositories, which can lead to communication delays, inconsistent records, and operational inefficiencies. To address these challenges, Paidi (2025) developed a Web-Based Real Estate Management System (REMS) supported by a structured Database Management System (DBMS) architecture designed to facilitate property listing management and streamline interactions between buyers and sellers.

The study emphasized that many real estate organizations continue to encounter difficulties in maintaining accurate and synchronized property information due to fragmented data storage practices. When information is distributed across multiple records or maintained manually, organizations become vulnerable to data inconsistencies, duplication of records, and delays in information retrieval. Recognizing these issues, the researcher proposed a centralized web-based platform that stores all property-related information within a relational database structure, ensuring that records remain consistent, organized, and easily accessible.

A major contribution of the study was the development of a specialized “Sale a Plot” module that allows administrators and authorized users to input and manage property information through structured forms. This module incorporated data validation mechanisms that verify the accuracy and completeness of information before records are stored within the database. Important property details such as registration numbers, location information, property dimensions, and pricing categories were validated at the point of entry to reduce errors and maintain data integrity. The researcher emphasized that validation procedures are essential for preventing inaccurate records from entering the system and compromising organizational operations.

The architecture of the proposed system was based on a three-tier framework consisting of the presentation layer, application layer, and database layer. This architectural approach separates user interface functions from business logic and data management processes. According to the study, the separation of responsibilities among system layers improves maintainability, scalability, and overall system performance. By isolating database operations from the user interface, the system was able to process requests more efficiently while maintaining a structured and organized codebase.

The researcher also evaluated the performance of the developed system through empirical testing. The results indicated that the platform achieved query response times of less than 300 milliseconds, demonstrating its ability to retrieve and process information rapidly. This level of responsiveness is particularly important in real estate environments where users frequently access property records, update information, and perform search operations. The study concluded that a well-designed relational database combined with a three-tier architecture significantly improves both system performance and data consistency.

Furthermore, the research highlighted the importance of centralized database management in supporting effective communication between stakeholders. By ensuring that all users access the same updated information from a single source, the system minimized misunderstandings, reduced duplicate records, and improved overall operational transparency. The researcher concluded that database-driven web applications provide an effective solution for modern real estate organizations seeking to improve information management and service delivery.

The findings of the study demonstrated that web-based real estate management systems supported by structured database architectures can effectively address many of the challenges associated with traditional property management. Through centralized data storage, validation mechanisms, and optimized system architecture, organizations can achieve greater efficiency, improved data accuracy, and enhanced user satisfaction.

Relevance to the Current Study

The study conducted by Paidi (2025) is highly relevant to the present research because it directly addresses many of the operational challenges currently experienced by Golden Dragon Estate Corporation. Similar to the conditions described in the study, the company relies on manually maintained records and physical subdivision maps to monitor lot availability. These traditional methods create risks related to information accuracy, record maintenance, and timely updates.

Paidi's findings provide strong evidence that centralized database-driven systems can successfully eliminate these limitations through digital transformation.

One of the most significant contributions of the study to the current research is its emphasis on structured database management and data validation. The proposed Web-Based Lot Availability Monitoring System similarly relies on a MySQL database to store and manage lot information. Through proper validation mechanisms, the proposed system can ensure that critical lot data—including lot numbers, block numbers, project locations, and availability statuses—remain accurate and consistent. This supports the objective of minimizing human errors commonly associated with manual record-keeping processes.

The study also validates the use of a three-tier system architecture, which aligns with the technical design of the proposed capstone project. In the current system, React serves as the presentation layer, Node.js and Express.js function as the application layer, and MySQL operates as the database layer. The successful implementation of this architecture in Paidi's study demonstrates its effectiveness in supporting responsive user interfaces, efficient data processing, and reliable database management.

Additionally, the rapid query response times achieved by the developed system highlight the importance of efficient information retrieval in real estate applications. Similar performance considerations are relevant to the proposed lot availability monitoring system, where authorized personnel must quickly access, search, and update lot information. The findings indicate that centralized database systems can support real-time operations while maintaining data integrity and responsiveness.

Although Paidi's research focused on broader buyer-seller interactions and property listing management, while the current study specifically addresses lot availability monitoring, both

systems share common objectives related to centralized data management, information accuracy, operational efficiency, and digital record keeping. Therefore, the study provides strong theoretical and technical support for the development of the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation

2.2 Local Studies

#1. Development and Evaluation of an Advanced Property Management System Using Unified OOAD for Optimizing Real Estate Operations

Encarnacion et al. (2025) developed and evaluated an Advanced Property Management System (PMS) designed to improve the efficiency of real estate operations in Siargao Island, Philippines. The study was conducted in response to the growing complexity of managing property information brought about by the continuous expansion of real estate developments within the region. The researchers recognized that traditional methods of property management often result in delayed information retrieval, inefficient record handling, and difficulties in monitoring property transactions. To address these challenges, they proposed a web-based system capable of centralizing property information and providing users with efficient access to real-time data.

The researchers utilized the Unified Process for Object-Oriented Analysis and Design (OOAD) as the primary software development approach in constructing the system. Through this methodology, they developed a multi-functional platform equipped with property directories, advanced search capabilities, status filtering mechanisms, and a centralized database architecture. The database served as the core component of the system, enabling users to store, retrieve, update, and manage property information efficiently. By integrating these functionalities into a single platform, the system provided a more organized and streamlined approach to managing real estate records.

To evaluate the effectiveness of the developed system, Encarnacion et al. (2025) employed the ISO/IEC 25010 Software Quality Standards. The evaluation involved information technology experts and prospective users who assessed the system based on software quality characteristics such as functional suitability, security, usability, reliability, and performance efficiency. The results revealed a high level of acceptance, achieving an overall mean rating of 4.52 out of 5.00. Notably, the system obtained ratings of 4.48 for functional suitability and 4.53 for security, indicating that the application successfully met user requirements while maintaining adequate protection of sensitive information.

The findings of the study demonstrated that web-based property management systems can significantly improve operational efficiency by centralizing records, automating information management processes, and providing real-time access to property data. Furthermore, the study emphasized the importance of implementing strong security measures and reliable database architectures to ensure data integrity and protect organizational information. The researchers concluded that digital property management platforms offer a practical and effective solution for addressing the increasing demands of modern real estate operations.

Relevance to the Current Study

The study conducted by Encarnacion et al. (2025) is highly relevant to the present study because both projects focus on improving real estate operations through the implementation of web-based information systems. Similar to the Property Management System developed in the study, the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation aims to replace manual processes with a centralized digital platform that improves information accessibility, operational efficiency, and data accuracy.

Both studies utilize database-driven architectures to manage and organize real estate information. While Encarnacion et al. focused on property directories, search functionalities, and property management operations, the current study specifically concentrates on monitoring residential lot availability and updating lot statuses in real time. Despite these differences in scope, both systems share the common objective of improving record management and reducing reliance on manual documentation.

Additionally, the use of ISO/IEC 25010 as an evaluation framework provides valuable guidance for assessing the quality of the proposed system. The study's positive results in functional suitability and security further support the importance of implementing reliable database management and access control mechanisms within the proposed lot monitoring system. Therefore, the study serves as a strong local reference that validates the effectiveness web-based solutions in enhancing real estate management operations within the Philippine setting.

#2. Web-Based Inventory Management System

Tanaman et al. (2023) developed a Web-Based Inventory Management System to address the operational challenges experienced by a commercial enterprise with multiple branches in Pagadian City, Zamboanga del Sur. The study was initiated in response to the organization's continued reliance on manual, paper-based inventory tracking processes, which resulted in delayed reporting, data inconsistencies, and difficulties in monitoring inventory records across different business locations. The researchers recognized that the traditional approach to inventory management was inefficient and vulnerable to human errors, making it difficult for management to obtain accurate and timely information regarding organizational assets.

To overcome these challenges, the researchers designed and implemented a web-based inventory management platform that centralized inventory records and automated the recording and reporting processes. The system enabled users to store, retrieve, and update inventory information electronically, eliminating the need for manual paper logs. Through the implementation of a digital database environment, inventory transactions could be processed more efficiently, while management personnel gained easier access to updated information through a centralized online platform.

One of the primary objectives of the study was to improve the accuracy and reliability of inventory records. By replacing manual documentation with automated data processing, the system significantly reduced the occurrence of recording errors and duplicated entries. The centralized database also ensured that information remained consistent across different branches, allowing authorized users to access the same records regardless of location. This capability improved organizational coordination and supported more effective decision-making processes.

The findings of the study revealed that the implementation of the web-based system resulted in substantial improvements in operational efficiency and record management. The researchers observed that inventory reports could be generated more quickly, information retrieval became significantly faster, and overall data accuracy improved. Furthermore, the digital platform provided a secure environment for managing organizational records while reducing the administrative burden associated with maintaining physical documents. The researchers concluded that web-based inventory systems offer a practical solution for organizations seeking to modernize traditional record-keeping processes and enhance operational productivity.

Relevance to the Current Study

The study conducted by Tanaman et al. (2023) is highly relevant to the present study because both projects address the limitations of manual, paper-based monitoring systems through the implementation of web-based information management solutions. Similar to the inventory management challenges identified in the study, Golden Dragon Estate Corporation currently relies on manually maintained subdivision layout maps to monitor lot availability. This traditional approach creates risks related to record inaccuracies, delayed updates, and difficulties in accessing current information.

Both studies emphasize the importance of centralized database management in improving operational efficiency and information accuracy. While the system developed by Tanaman et al. focuses on inventory tracking and asset management, the proposed Web-Based Lot Availability Monitoring System concentrates on monitoring residential lot statuses and maintaining updated lot records. Despite the difference in application domains, both systems share the common objective of replacing manual processes with automated digital solutions that provide real-time access to information.

Furthermore, the successful implementation of a centralized web-based platform in the study demonstrates the effectiveness of electronic record management in reducing human errors and improving organizational workflows. These findings support the development of the proposed system, which seeks to modernize lot monitoring operations by replacing physical subdivision maps with a secure and accessible digital database. Therefore, the study serves as a valuable local reference that validates the use of web-based information systems for improving record management, data accuracy, and operational efficiency.

#3. Development of GSO Item Borrowing and Inventory Management System

Estrella et al. (2026) developed a GSO Item Borrowing and Inventory Management System for the General Services Office (GSO) of Tagoloan, Misamis Oriental. The study was conducted to address inefficiencies associated with traditional paper-based processes used in tracking borrowed items, managing inventory records, and processing approval requests. The researchers observed that manual documentation often resulted in delayed transactions, inaccurate records, and difficulties in monitoring the status of organizational resources. These challenges prompted the development of a web-based system capable of digitizing inventory operations and improving the overall efficiency of resource management within the organization.

To modernize the existing workflow, the researchers designed and implemented a web-based inventory management platform supported by a relational MySQL database. The system automated various processes, including item borrowing requests, inventory tracking, approval workflows, and record maintenance. By replacing paper-based forms with digital transactions, the application enabled faster processing of requests and improved the accuracy of stored information. The centralized database architecture ensured that inventory records remained organized, accessible, and consistent across different users and departments.

A significant aspect of the study was the evaluation of both the technical performance and usability of the developed system. From a technical standpoint, the application successfully improved processing speed, enhanced data accuracy, and streamlined inventory management procedures. The implementation of modern web technologies and database management tools contributed to the reliability and effectiveness of the system in handling organizational records. As a result, users were able to access information more efficiently and perform inventory-related tasks with reduced administrative effort.

However, despite the system's strong technical performance, the researchers found that user acceptance was affected by interface design considerations. Using the System Usability Scale (SUS), the study revealed that users experienced some difficulty adapting to the new digital environment because the interface differed significantly from the paper-based workflows they had become accustomed to. The researchers concluded that successful implementation of information systems requires not only robust technical functionality but also user-centered interface design that reflects familiar operational practices. They emphasized that system usability plays a crucial role in ensuring successful adoption and long-term utilization within organizations.

The study ultimately demonstrated that web-based inventory and tracking systems can improve operational efficiency, record accuracy, and information accessibility. At the same time, it highlighted the importance of designing intuitive interfaces that align with user expectations and existing workflows. The researchers recommended that future systems prioritize both technical performance and usability to maximize organizational acceptance and effectiveness.

Relevance to the Current Study

The study conducted by Estrella et al. (2026) is relevant to the present study because both projects focus on replacing traditional paper-based tracking processes with web-based information systems supported by a centralized MySQL database. Similar to the inventory management challenges identified in the study, Golden Dragon Estate Corporation currently relies on manually maintained subdivision maps and physical records to monitor lot availability.

Both systems aim to improve information accuracy, streamline monitoring activities, and enhance operational efficiency through digital transformation.

The study also provides valuable insights regarding system usability and user acceptance. While the proposed Web-Based Lot Availability Monitoring System focuses on residential lot monitoring rather than inventory management, both systems require users to transition from familiar manual processes to a computerized environment. The findings of Estrella et al. emphasize the importance of designing interfaces that closely resemble existing workflows to minimize resistance to change and encourage user adoption.

Furthermore, the successful implementation of a MySQL-backed web application supports the technical architecture of the current study. By incorporating an interactive React-based dashboard that visually represents lot statuses through clear indicators and user-friendly controls, the proposed system seeks to apply the lessons identified in the study regarding usability and interface design. Therefore, the research serves as an important local reference that supports both the technical and user-centered aspects of the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation.

#4. PhilHealth Regional Office VIII Supply and Property Inventory Management System

Funcion et al. (2019) developed a web-based Supply and Property Inventory Management System for PhilHealth Regional Office VIII to improve the efficiency of property monitoring and inventory management within the organization. The study was conducted in response to the limitations of the institution's existing manual inventory procedures, which relied heavily on paper-based records and physical documentation. These traditional methods often resulted in delayed reporting, mathematical errors, inconsistencies in inventory records, and difficulties in tracking the movement of organizational assets across various departments. Recognizing these challenges, the researchers proposed an automated system capable of centralizing property information and providing real-time monitoring capabilities.

The developed system was designed to manage the recording, tracking, and monitoring of supplies and properties within the organization. Through the implementation of a centralized database and web-based platform, authorized personnel were able to access updated inventory information more efficiently. The system automated various inventory-related tasks, including property recording, status monitoring, report generation, and inventory verification. By replacing manual record-keeping procedures, the application reduced the time and effort required to perform routine inventory management activities.

One of the primary objectives of the study was to improve data accuracy and minimize human errors commonly associated with manual inventory processes. The centralized database ensured that property records remained consistent and accessible, allowing users to retrieve information quickly whenever needed. Additionally, the system enabled real-time tracking of organizational assets, providing management with continuous visibility regarding the availability, location, and status of properties under their supervision. This capability significantly improved inventory control and facilitated more informed decision-making processes.

The implementation of the system produced several positive outcomes for the organization. The researchers observed improvements in operational efficiency, data accuracy, and record accessibility. Manual counting procedures and repetitive documentation tasks were substantially reduced, enabling personnel to focus on more critical administrative responsibilities. Furthermore, the system enhanced accountability by providing a structured and reliable mechanism for monitoring property transactions and inventory movements. The researchers concluded that web-based inventory management systems are effective tools for improving organizational control, reducing operational risks, and modernizing traditional inventory practices.

The study emphasized the importance of adopting digital technologies to support efficient property monitoring and information management. By centralizing records and automating inventory processes, organizations can improve data reliability, accelerate information retrieval, and reduce the likelihood of errors associated with manual record-keeping systems.

Relevance to the Current Study

The study conducted by Funcion et al. (2019) is relevant to the present study because both projects seek to replace manual monitoring processes with centralized web-based information systems. Similar to PhilHealth Regional Office VIII, Golden Dragon Estate Corporation currently relies on manually maintained records and physical subdivision layout maps to monitor lot availability. These traditional methods present challenges related to information accuracy, record maintenance, and timely updates, making digital transformation a practical solution.

Both studies emphasize the importance of real-time monitoring and centralized database management. While the inventory management system developed by Funcion et al. focuses on tracking organizational properties and supplies, the proposed Web-Based Lot Availability Monitoring System concentrates on monitoring residential lot statuses such as available, pending, and sold. Despite the difference in application domains, both systems share the common objective of improving record accuracy, reducing manual workloads, and providing immediate access to updated information.

Furthermore, the successful implementation of automated status tracking in the study demonstrates the effectiveness of web-based systems in minimizing human errors and improving operational efficiency. These findings support the development of the proposed system, which aims to eliminate the limitations of manual lot monitoring by providing authorized personnel with a centralized platform for updating and viewing lot availability information. Therefore, the study serves as a valuable local reference that validates the use of web-based monitoring systems for improving inventory control, information management, and organizational efficiency.

#5. Re:SCN: A Web-Based Report Management System with RFID-Based Vehicle Monitoring for Villa Belen

Canlas et al. (2022) developed Re, a web-based report management and monitoring system integrated with RFID-based vehicle monitoring for Villa Belen Subdivision in Angeles City, Pampanga. The study was conducted to address the operational challenges experienced by residential communities that continue to rely on traditional administrative procedures and manual record-keeping practices. The researchers observed that the absence of a centralized digital platform often resulted in communication delays, inefficient monitoring processes, and difficulties in managing community-related records and activities. These limitations affected the overall efficiency of subdivision administration and hindered timely responses to operational concerns.

To address these challenges, the researchers designed and implemented a web-based system capable of consolidating administrative records, monitoring activities, and organizing information within a centralized platform. The system integrated reporting functionalities with RFID-based vehicle monitoring technology to improve the tracking and management of subdivision-related activities. Through a centralized database and web-accessible interface, administrators were able to access updated records, monitor community operations, and manage reports more efficiently. This digital approach reduced the reliance on manual documentation and improved the accessibility of information for authorized users.

One of the primary goals of the study was to enhance communication and streamline administrative processes within the residential community. The researchers found that the implementation of a centralized web-based platform significantly improved the organization of records and facilitated faster dissemination of information. Administrative personnel were able to monitor activities more effectively, generate reports with greater efficiency, and respond to concerns in a more timely manner. The system also reduced the confusion commonly associated with paper-based records by ensuring that all relevant information was stored in a single, structured repository.

To evaluate the effectiveness of the developed system, the researchers gathered feedback from users using a standardized Likert-scale assessment. The results indicated a high level of user satisfaction, with respondents reporting improvements in operational efficiency, information accessibility, and administrative transparency. The researchers concluded that web-based monitoring and management systems can successfully address the limitations of traditional subdivision administration by providing a reliable and organized platform for managing records and monitoring community activities.

The study highlighted the growing importance of digital transformation within residential developments and emphasized the value of centralized information systems in improving operational effectiveness. By integrating monitoring capabilities with web-based management tools, organizations can achieve greater transparency, reduce administrative errors, and improve overall service delivery.

Relevance to the Current Study

The study conducted by Canlas et al. (2022) is relevant to the present study because both projects focus on improving administrative operations within residential developments through the use of web-based information systems. Similar to Villa Belen Subdivision, Golden Dragon Estate Corporation manages property-related information that requires efficient monitoring, accurate record management, and timely updates. Both studies recognize the limitations of manual processes and propose digital solutions to improve operational performance.

Another significant similarity is the use of a centralized web platform to organize and monitor important records. While Re focuses on report management and vehicle monitoring within a residential subdivision, the proposed Web-Based Lot Availability Monitoring System concentrates on monitoring residential lot statuses and maintaining accurate lot inventory records. Despite the difference in functionality, both systems seek to improve information accessibility, reduce administrative workload, and enhance organizational efficiency through centralized digital management.

Furthermore, the positive user feedback reported in the study demonstrates the effectiveness of web-based systems in reducing administrative errors and improving transparency. These findings support the objectives of the proposed system, which aims to replace manual subdivision layout maps with a centralized digital platform capable of providing real-time lot availability information. Therefore, the study serves as a valuable local reference that validates the application of web-based technologies in supporting efficient management operations within Philippine residential developments.

#6. Vehicle Tracking and Monitoring: A Security System for Exclusive Subdivision

Lino and Festijo (2019) developed a Vehicle Tracking and Monitoring System designed to enhance security and monitoring operations within an exclusive subdivision in the Philippines.

The study was conducted in response to the growing need for real-time monitoring solutions capable of providing immediate updates regarding activities occurring within residential communities. The researchers observed that traditional monitoring methods often relied on manual procedures and fragmented records, making it difficult for subdivision administrators to obtain timely and accurate information regarding vehicle movements and security-related activities. These limitations created operational inefficiencies and reduced the effectiveness of existing monitoring practices.

To address these challenges, the researchers designed a web-based monitoring platform that provided real-time visibility of activities occurring within the subdivision. The system enabled administrators to track and monitor vehicle movements through a centralized dashboard that continuously displayed updated information. All monitoring data were stored within a database system, allowing users to access historical records and review past activities whenever necessary. Through the implementation of a centralized web interface, the system improved information accessibility and provided a more organized approach to security management.

A major contribution of the study was its emphasis on real-time data visualization. The researchers highlighted the importance of presenting information through an interactive dashboard that allowed administrators to immediately identify ongoing activities and respond to potential issues more effectively. By continuously updating monitoring data, the system eliminated the delays associated with traditional reporting methods and improved situational awareness among subdivision personnel. This capability contributed to more efficient decision-making and enhanced operational control within the residential community.

The researchers evaluated the developed system based on several software quality criteria, including functionality, usability, and operational efficiency. The results indicated that the system performed effectively in delivering accurate and timely information while maintaining a user-friendly interface. Respondents reported that the centralized dashboard improved monitoring efficiency and provided a practical solution for managing subdivision security operations. The study concluded that web-based monitoring systems are highly effective tools for supporting real-time information management and improving administrative oversight within residential communities.

Furthermore, the study emphasized that modern subdivisions increasingly require digital monitoring platforms capable of providing continuous access to updated information. The successful implementation of the system demonstrated that web-based technologies can improve operational transparency, facilitate information sharing, and support more efficient management practices within specialized residential environments.

Relevance to the Current Study

The study conducted by Lino and Festijo (2019) is relevant to the present study because both projects focus on the development of web-based monitoring systems designed to provide real-time information within residential property environments. Similar to the Vehicle Tracking and Monitoring System, the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation seeks to provide administrators with immediate access to updated information through a centralized digital platform.

One of the most significant similarities between the two studies is the emphasis on real-time data visualization through interactive dashboards. While the system developed by Lino and Festijo monitors vehicle movements and security-related activities, the proposed system focuses on monitoring residential lot availability and tracking lot status changes. In both cases, users rely on centralized dashboards to access current information and make timely decisions based on updated records.

Additionally, the study supports the importance of maintaining a centralized database that stores both current and historical information for future reference. This concept aligns with the objectives of the proposed system, which aims to maintain accurate records of lot statuses while allowing authorized users to monitor updates efficiently. The positive evaluation results reported in the study further demonstrate the effectiveness of web-based monitoring platforms in improving operational efficiency and information accessibility.

Although the application domains differ, both studies address common challenges related to information monitoring, data accessibility, and administrative control. Therefore, the findings of Lino and Festijo (2019) provide valuable local support for the implementation of an interactive web dashboard and real-time monitoring capabilities within the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation.

#7. eReserve: An Online Facility Reservation System of One Philippine Higher Education Institution

Vega and Generoso (2025) developed eReserve, an online facility reservation system implemented in Lyceum of the Philippines University – Batangas to address inefficiencies in managing institutional facilities and resources. The study was conducted due to recurring operational issues such as scheduling conflicts, overlapping reservations, and delays in updating facility availability caused by reliance on manual reservation logs. These challenges negatively affected productivity and created confusion among users regarding the actual availability of institutional assets and spaces.

To resolve these issues, the researchers designed and implemented a web-based reservation platform supported by a MySQL database engine. The system was capable of performing real-time availability checks and automatically updating reservation statuses upon user requests. Through this centralized approach, the system ensured that all facility bookings were properly recorded, validated, and synchronized within the database. This eliminated the inconsistencies commonly associated with manual reservation systems and improved the reliability of scheduling processes.

A key feature of the system was its ability to perform real-time validation and automated conflict detection. When users attempted to reserve a facility, the system immediately checked availability and prevented double bookings by locking the selected asset once confirmed. This functionality significantly reduced administrative errors and ensured that facility schedules remained accurate and up to date. Additionally, the system provided detailed utilization records that allowed administrators to monitor usage patterns and optimize resource allocation.

The evaluation of the system demonstrated improvements in data accuracy, operational efficiency, and user satisfaction. The researchers found that replacing manual reservation processes with a centralized digital platform significantly reduced entry errors and improved the speed of processing reservations. The system also enhanced transparency by providing users with real-time access to facility availability, thereby reducing confusion and improving communication between administrators and users.

The study concluded that web-based reservation systems with real-time validation mechanisms are highly effective in improving resource management and operational efficiency in institutional settings. By centralizing data and automating availability checks, organizations can significantly reduce administrative workload while improving the accuracy and reliability of their operational processes.

Relevance to the Current Study

The study conducted by Vega and Generoso (2025) is relevant to the present study because both systems rely on real-time availability tracking and centralized database management to improve operational efficiency. Similar to the eReserve system, the proposed Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation aims to provide accurate, real-time updates on the status of assets—specifically residential lots categorized as available, pending, or sold.

One of the strongest similarities between the two studies is the use of automated availability verification to prevent conflicts and ensure data accuracy. While eReserve prevents double booking of institutional facilities, the proposed system ensures that residential lots cannot be mistakenly marked or updated by multiple users simultaneously. Both systems rely on MySQL-backed validation mechanisms to maintain consistency and integrity in recorded data.

Additionally, both studies emphasize the importance of centralized web-based platforms in eliminating manual tracking errors and improving user accessibility. The findings of Vega and Generoso demonstrate that real-time status updates significantly reduce administrative inefficiencies, which directly supports the objectives of the current study. By adopting a similar architecture, the proposed system enhances transparency and ensures that authorized personnel can reliably monitor and update lot availability information.

Although the eReserve system is designed for facility reservation in an academic institution, while the current study focuses on residential lot monitoring in a real estate environment, both share the same fundamental objective of managing asset availability through automated, database-driven systems. Therefore, the study provides strong local evidence supporting the effectiveness of real-time web-based monitoring and validation mechanisms in improving operational control and accuracy.

Summary of Local Studies

The reviewed local studies consistently demonstrate the positive impact of web-based information systems on organizational operations. Researchers have found that centralized databases, automated information processing, and digital monitoring tools improve efficiency, reduce human errors, and enhance data accessibility.

Although the reviewed studies focus on different application domains, they share common objectives related to information management, record monitoring, and operational improvement. These findings provide strong support for the development of the proposed Web-Based Lot Availability Monitoring System.

Furthermore, the reviewed local studies reveal a growing trend among Philippine organizations toward digital transformation and automation. This trend highlights the relevance and practicality of implementing a computerized lot monitoring system within Golden Dragon Estate Corporation.

2.3 Related Literature

Real-Time Inventory Automation vs. Manual Spreadsheet Tracking

Immadisetty (2025) examined the operational risks and inefficiencies associated with traditional inventory management systems, particularly those relying on manual files and spreadsheet-based tracking. The study emphasized that inventory management plays a critical role in asset-heavy organizations, where even small errors in tracking can lead to significant financial and operational losses. One of the key issues identified in the study is the presence of stock shortages and over-allocation, which are often caused by outdated or inaccurate records within manual tracking systems.

The author explained that traditional spreadsheet-based workflows are inherently limited due to their static nature. Since data is manually encoded and updated, these systems are highly vulnerable to human error, delayed updates, and inconsistencies across different records. This creates a condition known as structural data latency, where decision-makers are forced to rely on outdated information rather than real-time data. As a result, organizations experience difficulties in maintaining accurate visibility over their assets, which directly impacts operational efficiency and decision-making.

To address these limitations, Immadisetty (2025) proposed the use of real-time inventory automation systems that utilize centralized databases and continuous data synchronization. These systems allow for instant updates whenever transactions occur, ensuring that all users have access to the most current and accurate information. The study highlights that real-time analytics significantly improve managerial decision-making by providing continuous visibility into inventory levels, usage patterns, and asset distribution. This shift from static to dynamic data processing eliminates delays associated with manual reconciliation and reduces the risk of reporting inaccuracies.

Another key contribution of the study is its emphasis on automation as a mechanism for reducing human error. By eliminating manual encoding processes, automated systems ensure that data is captured accurately at the point of entry. This not only improves data integrity but also enables organizations to build reliable datasets that can support advanced analytics and forecasting. The study concludes that automation is essential for organizations seeking to transition from reactive to proactive asset management strategies.

Overall, Immadisetty (2025) demonstrates that real-time inventory systems provide a more reliable, efficient, and scalable solution compared to traditional spreadsheet-based methods. The findings strongly support the adoption of digital transformation initiatives in asset tracking environments where accuracy and timeliness are critical.

Relevance to the Current Study

The study conducted by Immadisetty (2025) is relevant to the present study because both emphasize the importance of replacing manual tracking systems with automated, real-time database-driven solutions. Similar to inventory management challenges discussed in the study, Golden Dragon Estate Corporation currently relies on manual and semi-digital methods, such as physical subdivision maps and paper-based tracking, to monitor lot availability. These traditional methods are prone to delays, inconsistencies, and human errors, particularly in updating lot statuses.

Both studies highlight the importance of real-time data accuracy in preventing operational issues such as double-booking, misclassification, and outdated records. While the literature focuses on general inventory systems, the proposed Web-Based Lot Availability Monitoring System applies the same principles to residential lot management by ensuring that updates to lot statuses are reflected instantly within a centralized database.

Furthermore, the study supports the technical direction of the proposed system, which aims to eliminate spreadsheet dependency and replace it with a structured web-based architecture. By implementing a real-time monitoring system with centralized data control, Golden Dragon Estate Corporation can improve operational efficiency, reduce human error, and ensure accurate lot status tracking. Therefore, this literature provides a strong conceptual foundation for the development of the proposed system.

Single Page Application (SPA) Architecture via React.js

Kothapalli (2022) discussed the importance of Single Page Application (SPA) architecture in modern web development, emphasizing its role in delivering fast, responsive, and seamless user experiences. The study explains that SPAs are designed to load a single HTML page and dynamically update content without requiring full page reloads. This approach significantly improves performance and user interaction, especially in systems that involve frequent data updates and real-time information display.

The author highlighted that SPA architecture is particularly effective in handling complex and data-intensive applications. Through client-side rendering, SPAs reduce server load and improve system responsiveness by processing user interactions directly within the browser. This allows users to navigate between different sections of a system without experiencing delays associated with traditional multi-page web applications. As a result, SPAs provide a smoother and more efficient user experience.

Kothapalli (2022) also identified key optimization techniques that enhance SPA performance, including code splitting, lazy loading, client-side caching, and centralized state management. These techniques help ensure that applications remain efficient even as they scale in size and complexity. Code splitting allows only necessary components to load when required, while lazy loading defers non-essential resources to improve initial load time. Meanwhile, client-side caching and state management contribute to maintaining consistent application behavior and reducing redundant server requests.

The study further explains that the effectiveness of SPA architecture depends on maintaining a balance between performance optimization and functional integrity. While advanced techniques can improve speed and responsiveness, they must be implemented carefully to ensure that the application's core functionality and user experience are not compromised. This principle is especially important in systems that require continuous interaction with dynamic data.

Overall, the study concludes that SPA architecture has become a dominant approach in modern web engineering due to its ability to deliver responsive, efficient, and scalable applications. It is particularly suitable for systems that require frequent updates and real-time data visualization.

Relevance to the Current Study

The study conducted by Kothapalli (2022) is relevant to the present study because it supports the use of React.js as the frontend framework for the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Similar to the SPA architecture discussed in the study, the proposed system requires a highly responsive interface that allows users to view and update lot statuses in real time without unnecessary page reloads.

Both the study and the proposed system emphasize the importance of handling dynamic data efficiently. In the context of lot monitoring, real estate administrators need to interact with frequently changing information such as lot availability, status updates, and visual layout representations. SPA architecture enables these interactions to occur smoothly by updating only specific components of the interface rather than reloading the entire page.

Furthermore, the optimization techniques discussed by Kothapalli (2022), such as client-side caching and centralized state management, directly support the development of a scalable and efficient system. These techniques align with the proposed system's goal of providing fast and reliable access to real-time data while maintaining system performance. Therefore, this literature provides strong technical justification for the adoption of React.js and SPA architecture in the development of the proposed system.

Server-Side Engineering with Node.js and Express.js

Shkodra et al. (2021) examined the performance and efficiency of modern backend development environments, with a particular focus on Node.js and its suitability for building scalable web applications. The study emphasized the increasing demand for backend systems capable of handling concurrent user requests efficiently, especially in applications that rely heavily on real-time data processing and frequent database interactions. Traditional server-side architectures often struggle with high concurrency due to blocking operations, which can lead to performance bottlenecks and reduced system responsiveness.

The researchers highlighted that Node.js addresses these limitations through its asynchronous, event-driven, and non-blocking Input/Output (I/O) model. This architecture allows the server to handle multiple client requests simultaneously without waiting for one process to complete before starting another. As a result, Node.js is particularly effective for applications that require real-time data exchange, such as web dashboards, APIs, and monitoring systems. The study further explains that this design significantly improves scalability and responsiveness in multi-user environments.

In addition to Node.js, the study also discussed the role of Express.js as a minimal and flexible web application framework that simplifies backend development. Express.js provides essential tools for building RESTful APIs, managing routes, and handling HTTP requests efficiently. When combined with Node.js, it enables developers to create structured and maintainable backend systems capable of supporting Create, Read, Update, and Delete (CRUD) operations. The study also noted that this combination performs effectively under typical enterprise workloads, particularly in systems that do not require extremely high computational intensity.

Shkodra et al. (2021) concluded that while more complex, compiled backend frameworks may outperform Node.js under extreme computational demands, Node.js and Express.js remain highly efficient for standard web-based applications. Their lightweight structure, ease of integration, and scalability make them suitable for modern software systems that prioritize speed, flexibility, and real-time responsiveness.

Relevance to the Current Study

The study conducted by Shkodra et al. (2021) is relevant to the present study because it provides strong technical justification for the use of Node.js and Express.js as the backend architecture for the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Similar to the systems described in the study, the proposed system requires a backend capable of handling multiple simultaneous user requests, particularly when real estate agents are updating or querying lot availability data.

Both the study and the proposed system rely on RESTful API communication to manage data transactions between the client and server. The asynchronous and non-blocking nature of Node.js ensures that concurrent operations—such as updating lot statuses, retrieving property records, and processing user authentication—can be executed efficiently without causing delays or server overload. This is especially important in real estate operations where real-time accuracy is essential for preventing conflicts such as double-booking or outdated status displays.

Furthermore, the combination of Node.js and Express.js supports scalable system development, allowing the proposed platform to grow in functionality as organizational needs evolve. The study confirms that this technology stack is well-suited for enterprise-level applications that require reliability, responsiveness, and efficient data processing. Therefore, this literature strongly supports the technical framework chosen for the development of the proposed system.

Relational Database Management Systems (RDBMS) via MySQL

Chittayasothorn (2022) discussed the foundational principles of Relational Database Management Systems (RDBMS), emphasizing the importance of structured data organization through relational models. The study explains that relational databases are built on the concept of relational algebra, where data is stored in structured tables consisting of rows and columns. This structure allows developers to organize complex datasets in a logical and consistent manner, making it easier to retrieve, update, and manage information efficiently.

The author highlighted that the use of Structured Query Language (SQL) plays a critical role in managing relational databases. SQL provides a standardized way of interacting with data, enabling operations such as insertion, deletion, updating, and querying of records. Through relational schemas, developers can define relationships between different data entities, ensuring that data remains consistent and well-structured across the system. This structured approach is particularly important in systems that require accurate and reliable data handling.

A key focus of the study is the distinction between relational database structure and transactional integrity. Chittayasothorn (2022) emphasized that simply using a relational database system does not automatically guarantee data reliability. Instead, ensuring data integrity requires careful implementation of database design principles such as normalization and proper schema planning. Additionally, the study highlights the importance of Atomicity, Consistency, Isolation, and Durability (ACID) properties in maintaining reliable transactions within a database system.

The author further explained that ACID properties must be intentionally enforced through proper database design and transaction management. Atomicity ensures that transactions are fully completed or fully reverted, consistency maintains valid data states, isolation prevents conflicts between concurrent transactions, and durability guarantees that committed changes are permanently stored. Without these safeguards, even a relational database system may become vulnerable to data inconsistencies and corruption.

Overall, the study concludes that relational database systems remain one of the most reliable and widely used approaches for structured data management, particularly when combined with proper schema design and transaction control mechanisms. These principles are essential for maintaining long-term data integrity and system reliability in any data-driven application.

Relevance to the Current Study

The study conducted by Chittayasothorn (2022) is relevant to the present study because it provides a strong theoretical foundation for the use of MySQL as the relational database management system in the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Similar to the principles discussed in the study, the proposed system requires a highly structured and reliable database to manage critical information such as lot availability, status changes, and property allocations.

Both the study and the proposed system emphasize the importance of data integrity and consistency. In the context of real estate management, accurate tracking of lot statuses is essential to prevent errors such as duplicate allocations, incorrect status updates, and inconsistent records. The application of relational database principles ensures that all data relationships are properly defined and maintained within the system.

Furthermore, the emphasis on ACID properties directly supports the reliability requirements of the proposed system. Since lot status changes involve important operational decisions, it is critical that every transaction is processed accurately and securely. By implementing a well-designed relational schema with enforced constraints, the system ensures that all updates are properly recorded and protected from data corruption. Therefore, this literature provides essential technical justification for the adoption of MySQL and structured database design in the development of the proposed system.

Software Usability and Evaluation Quality Metrics

Klug (2017) presented a methodological overview of the System Usability Scale (SUS), originally developed by John Brooke in 1986, as a widely adopted tool for evaluating software usability. The study explains that SUS is a technology-independent, ten-item Likert-scale questionnaire designed to measure users' subjective perceptions of a system's usability. It assesses key aspects such as ease of use, system complexity, learnability, and overall user satisfaction. Because of its simplicity and reliability, SUS has become one of the most commonly used instruments for evaluating user experience in web-based systems and interactive applications.

The study emphasizes that SUS is particularly valuable in assessing systems intended for non-technical users, as it captures real user feedback in a standardized and quantifiable manner. By converting subjective opinions into numerical scores, SUS allows researchers and developers to compare usability levels across different systems or iterations. This makes it an effective tool for identifying usability issues and guiding interface improvements during system development.

In addition to usability measurement, Awangditama et al. (2023) discussed the importance of using standardized software quality models, particularly the ISO/IEC 25010 framework, for evaluating system performance and compliance. The framework provides a structured approach for assessing software quality across multiple dimensions, including functional suitability, performance efficiency, usability, reliability, security, and maintainability. This allows developers and evaluators to ensure that a system meets both technical and user-oriented requirements.

The study further explains that ISO/IEC 25010 serves as an objective evaluation framework that complements user-based assessment tools such as SUS. While SUS focuses on user satisfaction and interface experience, ISO/IEC 25010 provides measurable technical criteria that can be used to assess system performance and structural quality. By combining both approaches, developers can achieve a more comprehensive evaluation of software systems, ensuring that both usability and technical integrity are properly addressed.

Overall, these evaluation methodologies highlight the importance of balancing user experience with technical performance in software development. The integration of both subjective and objective evaluation tools ensures that systems are not only functional but also efficient, reliable, and user-friendly.

Relevance to the Current Study

The study conducted by Klug (2017) and Awangditama et al. (2023) is relevant to the present study because it provides essential evaluation frameworks that will be used in assessing the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Similar to the proposed system, modern web-based applications require both usability testing and technical evaluation to ensure overall effectiveness and user satisfaction.

Both SUS and ISO/IEC 25010 will be applied in the evaluation of the proposed system. The SUS framework will be used to measure the perceived usability of the system among end-users such as real estate personnel who are responsible for monitoring and updating lot statuses. This is particularly important because the success of the system depends on how easily users can transition from manual, color-coded lot maps to a digital platform.

Meanwhile, the ISO/IEC 25010 model will be used to assess the technical quality of the system, including its functionality, performance efficiency, reliability, and security. Since the system handles critical real estate data such as lot availability and status updates, it is essential to ensure that it performs accurately and securely under operational conditions. Therefore, these frameworks provide a comprehensive evaluation strategy that supports both the usability and technical validation of the proposed system.

Role-Based Access Control (RBAC)

Owoade et al. (2023) discussed Role-Based Access Control (RBAC) as a fundamental security model used in modern enterprise systems to manage user authentication and authorization. The study explains that RBAC is a structured approach where system permissions are assigned based on predefined organizational roles rather than individual user identities. This model allows system administrators to efficiently control access rights by grouping users according to their job functions and responsibilities within the organization.

The researchers emphasized that RBAC is particularly effective in multi-user environments where different levels of access are required. By assigning roles such as administrator, manager, or standard user, the system can ensure that each user only has access to the functions necessary for their tasks. This minimizes the risk of unauthorized access to sensitive information and reduces the likelihood of accidental or intentional data modification. As a result, RBAC enhances both system security and operational efficiency.

Another key advantage of RBAC highlighted in the study is its scalability and ease of management. As organizations grow and new users are added, administrators can simply assign existing roles rather than configuring permissions individually for each user. This reduces administrative overhead and ensures consistent enforcement of security policies across the system. Additionally, RBAC supports compliance with organizational security standards by providing clear documentation of access control structures.

The study further explains that RBAC plays a critical role in protecting sensitive data within enterprise applications. By restricting access to critical system functions—such as data modification, deletion, or administrative configuration—RBAC ensures that only authorized personnel can perform high-level operations. This layered security approach helps maintain data integrity and prevents unauthorized system changes.

Overall, Owoade et al. (2023) conclude that RBAC is an essential security mechanism for modern web-based systems, particularly those handling sensitive or operationally critical data. Its structured approach to access control makes it a reliable and scalable solution for managing user permissions in complex applications.

Relevance to the Current Study

The study conducted by Owoade et al. (2023) is relevant to the present study because it provides a strong security framework for implementing Role-Based Access Control in the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Similar to the RBAC model described in the study, the proposed system requires strict access control to ensure that only authorized personnel can modify sensitive lot information.

Both studies emphasize the importance of separating user roles to protect data integrity. In the proposed system, real estate agents may be granted limited access, such as viewing lot availability and performing basic searches, while administrators are given full control over critical functions such as updating lot statuses (available, pending, or sold). This role separation ensures that unauthorized modifications are prevented and that the system maintains accurate and reliable records.

Furthermore, the scalability of RBAC supports the long-term use of the system as the organization grows or introduces new user roles. The structured permission system reduces administrative complexity and ensures consistent enforcement of security policies across all users. Therefore, this literature provides essential security justification for incorporating RBAC into the design and implementation of the proposed system.

Agile Software Development Framework

Irfansyah et al. (2025) discussed the importance of Agile software development methodologies, particularly the Scrum framework, in managing complex and evolving software projects. The study highlights that modern software systems, especially web-based applications, require flexible and iterative development approaches due to continuously changing user requirements and business environments. Traditional linear development models often struggle to accommodate these changes, leading to delays, misalignment with user needs, and reduced system effectiveness.

The researchers explained that Scrum, as an Agile framework, addresses these challenges by dividing the development process into smaller, manageable cycles known as sprints. Each sprint focuses on the incremental delivery of system components, such as user authentication modules, data processing features, or interface elements. This iterative approach allows development teams to continuously refine the system based on feedback gathered after each sprint, ensuring that the final product aligns closely with user expectations and operational requirements.

A key advantage of the Agile methodology is its emphasis on collaboration and adaptability. Irfansyah et al. (2025) emphasized that frequent communication between developers, stakeholders, and end-users is essential for identifying issues early in the development process. By incorporating regular feedback loops, teams can make timely adjustments to system design, functionality, and user interface components. This reduces the risk of major system redesigns at later stages and improves overall project efficiency.

The study further explains that Agile methodologies are particularly effective for user-centered applications where requirements may evolve throughout the development lifecycle. The flexibility of iterative development ensures that systems remain responsive to real-world needs, especially in environments where operational workflows are dynamic and subject to change. As a result, Agile frameworks enhance both product quality and user satisfaction.

Overall, the study concludes that Agile development, particularly Scrum, provides a structured yet flexible approach to software engineering. It enables teams to deliver functional software incrementally while continuously improving system quality through iterative refinement and stakeholder involvement.

Relevance to the Current Study

The study conducted by Irfansyah et al. (2025) is relevant to the present study because it supports the use of the Agile methodology in developing the Web-Based Lot Availability

Monitoring System for Golden Dragon Estate Corporation. Similar to the systems discussed in the study, the proposed application requires continuous refinement based on user feedback to ensure that it aligns with the actual workflows of real estate personnel.

Both studies emphasize the importance of iterative development and stakeholder collaboration. By dividing the development process into sprints, the proposed system can be progressively built, tested, and improved, ensuring that each module—such as lot status management, user authentication, and dashboard visualization—is properly validated before full deployment. This reduces development risks and ensures higher system reliability.

Furthermore, Agile methodology supports adaptability, which is essential for the proposed system since operational requirements may evolve during implementation. As Golden Dragon Estate Corporation transitions from manual color-coded maps to a digital monitoring system, user feedback will play a critical role in shaping interface design and functionality. Therefore, this literature provides strong methodological justification for adopting Agile Scrum in the development of the proposed system.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign studies, local studies, and related literature collectively present a strong foundation for understanding the development and implementation of modern web-based information systems. Across all sources, there is a consistent emphasis on the transition from manual, paper-based processes to automated, database-driven systems that improve operational efficiency, accuracy, and accessibility of information. Most of the studies highlight that traditional methods of record-keeping, such as physical documents and spreadsheet-based tracking, are prone to human error, data inconsistency, delayed updates, and inefficiencies in data retrieval.

Foreign studies primarily focus on advanced architectural implementations of real estate and inventory management systems using modern web technologies such as React.js, Node.js, Express.js, and MySQL. These studies emphasize the importance of centralized databases, real-time updates, role-based access control, and responsive user interfaces. They also highlight the significance of system security and scalability in handling concurrent users and dynamic data environments. Furthermore, foreign literature consistently supports the use of Single Page Application (SPA) architecture to enhance user experience and system responsiveness.

Local studies, on the other hand, provide contextual evidence within the Philippine setting, particularly in government offices, subdivisions, and local enterprises. These studies reinforce the effectiveness of web-based systems in replacing manual workflows and improving operational efficiency in real-world environments. They also emphasize usability concerns, particularly the importance of aligning system interfaces with existing user workflows to ensure smooth adoption. In addition, local studies highlight that while technical performance is important, user acceptance and interface familiarity play a critical role in the success of system implementation.

The related literature further strengthens the conceptual and technical foundation of the study by discussing key system components such as real-time inventory automation, backend server performance, relational database management, Agile methodology, security frameworks, and software evaluation metrics. These concepts collectively provide the theoretical basis for designing a reliable, secure, and efficient web-based system capable of handling real-time data updates and multi-user interactions.

Overall, the synthesis of literature reveals that modern web-based systems are most effective when they integrate four key components: (1) centralized database architecture for data integrity, (2) real-time processing for accurate updates, (3) secure role-based access control for system protection, and (4) user-centered interface design for usability and adoption. These combined elements form the essential foundation for building efficient and scalable information systems.

Research Gap

Despite the extensive development of web-based inventory, property management, and monitoring systems in both foreign and local contexts, there remains a specific gap in the application of these technologies to residential lot availability monitoring using a visual, map-based, and status-driven interface tailored for real estate operations in local Philippine settings.

Most existing studies focus on generalized domains such as inventory management, facility reservation, property listings, or asset tracking systems. While these systems successfully demonstrate the benefits of centralized databases, real-time updates, and role-based access control, they do not fully address the unique operational requirements of real estate companies that rely on spatial and visual representations of land parcels. In particular, the manual use of color-coded subdivision maps remains a common practice among local real estate agencies, despite its inefficiency and vulnerability to human error and data loss.

Furthermore, while many studies implement real-time tracking and database systems, few integrate all critical components—such as SPA-based interactive dashboards, RBAC security models, Agile development methodology, and real-time lot status visualization—into a single cohesive system specifically designed for real estate lot monitoring. This fragmentation of system capabilities highlights the absence of a unified solution that directly addresses the operational workflow of companies like Golden Dragon Estate Corporation.

Therefore, there is a clear need for a centralized, secure, and real-time web-based lot availability monitoring system that integrates modern web technologies with intuitive visual mapping interfaces. This system aims to bridge the gap by providing an efficient alternative to manual tracking methods, ensuring accurate lot status updates, reducing human error, and improving overall operational efficiency within real estate management processes.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

The study employed a developmental research design to design, develop, and evaluate the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation. Developmental research is a systematic process of creating and improving a product, system, or tool intended to address a specific real-world problem.

This research design is appropriate for the study because the primary objective is not only to analyze existing problems but also to produce a functional and usable web-based system that will replace the company's current manual lot monitoring process. The system aims to improve efficiency, accuracy, accessibility, and security in managing lot availability information.

The developmental process involves identifying existing operational issues, gathering requirements, designing system solutions, developing the system using appropriate technologies, and evaluating its performance based on user needs. Through this structured approach, the researchers ensure that the final output aligns with the actual requirements of Golden Dragon Estate Corporation.

Furthermore, the research design allows iterative improvements throughout development. Feedback from users and stakeholders is incorporated to refine system functionalities and ensure usability. This makes the approach suitable for Agile development, where continuous improvement is a core principle.

3.2 System Development Methodology

The development of the system follows the Agile Software Development Methodology. Agile is an iterative and incremental approach to software development that emphasizes flexibility, collaboration, and continuous improvement. Unlike traditional methodologies, Agile allows requirements and solutions to evolve through continuous feedback from users.

The Agile approach is suitable for this study because the requirements of the real estate company may change as the system is developed and tested. It also allows the development team to deliver functional modules in stages, ensuring that each part of the system is properly evaluated before full deployment.

The Agile methodology used in this study consists of the following phases:

3.2.1 Planning Phase

The Planning Phase involves identifying the overall objectives, scope, and requirements of the system. During this phase, the researchers conducted initial discussions and observations at

Golden Dragon Estate Corporation to understand the current manual process of monitoring lot availability.

It was found that the company uses printed subdivision maps where lot statuses are manually color-coded as green (available), yellow (pending), and red (sold). This method is time-consuming and prone to errors.

Based on these observations, the researchers defined the system goals, including:

- Automating lot status updates
- Providing an interactive lot monitoring interface
- Centralizing data storage
- Securing access through authentication
- Improving accessibility and efficiency

The technologies were also selected during this phase:

- React for frontend development

- Node.js and Express.js for backend development
- MySQL for database management
- Tailwind CSS for UI styling
- Nodemailer SMTP for email notifications

3.2.2 Analysis Phase

The Analysis Phase focuses on understanding the system requirements in detail. The researchers examined how information flows within the organization and identified the limitations of the current manual system.

The analysis revealed several issues:

- Delayed updating of lot status information
- Risk of human error in manual encoding
- Difficulty in tracking lot availability in real time
- Lack of centralized data storage
- Vulnerability of physical maps to damage or loss

From these findings, both **functional and non-functional requirements** were identified.

Functional requirements include:

- User authentication (login/logout)
- Lot status updating

- Interactive lot viewing system
- User Management
-
- Data storage and retrieval
- Email notifications

None-functional requirements include :

- Security data and user access
- System performance and responsiveness
- Ease of user (usability)
- System reliability and availability
- Data integrity and accuracy

3.2.3 Design Phase

The Design Phase involves creating the blueprint of the system. This includes planning how the system will function, how data will be stored, and how users will interact with it.

During this phase, the researchers developed :

- System architecture design
- Database Schema
- Entity-Relationship Diagram (ERD)
- Data Flow Diagrams (DFD)
- Use Case Diagrams

- User interface mockups

The system was designed as a **web-based client-server application**, where:

- The frontend (React) handles user interaction

- The backend (Node. js + Express) processes requests
- The database (MySQL) stores all system data

The interface design focuses on simplicity and usability. The main feature is the interactive lot monitoring map, which uses color-coded indicators to represent lot status:

- Green = Available
- Yellow -Pending
- Red - Sold

This design allows users to quickly interpret lot status without reading data tables.

3.2.4 Development and Implementation Phase

The Development Phase is where the actual system is built. The Agile approach allows development to be done in iterations, where each sprint focuses on specific modules.

The system was developed in the following modules:

- Authentication Module
- Lot Management Module
- User Management Module
- Dashboard Module
- Email Notification Module
- Reporting Module

Frontend development was done using **React and Tailwind CSS**, ensuring responsiveness and a clean user interface. Backend development used **Node.js and Express.js** to handle server-side logic and API requests. **MySQL** was used as the database to store users, lot data, and transaction records.

Each module was tested after development before moving to the next iteration. Feedback from users and group members was used to improve functionality and usability.

3.2.5 Testing Phase

The Testing Phase ensures that the system functions correctly and meets the required specifications. Various testing methods were applied:

- Functional testing to verify system features
- Integration testing to ensure modules work together
- Interface testing to evaluate interaction
- Security testing to check authentication and access control

Errors and bugs discovered during testing were corrected immediately. This iterative testing approach is aligned with Agile methodology, where testing is continuous throughout development.

3.2.6 Deployment and Maintenance Phase

The Deployment Phase involves releasing the system for actual use by Golden Dragon Estate Corporation. The system is installed on a server where authorized users can access it through a web browser.

User accounts are created for employees with different roles and access levels. Training is also provided to ensure users understand how to operate the system.

Maintenance includes:

- Bug fixing
- Performance optimization
- Security updates
- Feature enhancements based on user feedback

This ensures the system remains functional and relevant over time.

3.3 System Architecture

The system follows a **client-server architecture**, which is commonly used in modern web-based applications. This architecture separates the system into two main components: the client (frontend) and the server (backend), with a database layer supporting data storage.

3.3.1 Overview of the System

The Web-Based Lot Availability Monitoring System is designed to allow authorized users to view, update, and manage lot availability in real time. The system replaces the manual process of using printed subdivision maps with a digital and interactive platform.

When a user interacts with the system (e.g., updating lot status), the request is sent from the client-side interface to the server. The server processes the request, interacts with the database, and returns the updated information to the client.

3.3.2 Client-Server Setup

The system is composed of the following components:

Client Side (Frontend)

- Built using React
- Handles user interface and interactions
- Displays lot maps, dashboards, and forms

Server Side (Backend)

- Built using Node. js and Express. Js
- Handles API request
- Processes business logic
- Manages authentication and authorization

Database Layer

- MySQL database
- Stores user data, lot information, and status updates

3.3.3 User Interaction Flow

1. The user logs in using credentials
2. The system verifies authentication through the server
3. The user access the dashboard
4. The system displays the interactive lot map
5. The users selects a lot
6. The user updates the lot status (available, pending, sold)
7. The sever processes the requests
8. The database is updated
9. The system refreshes the interface with updated information
10. Email notification are sent if required

3.3 Tools and Technology Used

The development of the Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation utilized a combination of modern web development tools, frameworks, programming languages, and database technologies. These tools were selected to ensure that the system is efficient, scalable, secure, and user-friendly. The chosen technology stack supports both front-end and back-end development, allowing for a complete full-stack web application.

3.4.1 Programming Languages

Javascript

JavaScript is the primary programming language used in the development of the proposed system. It is a high-level, dynamic programming language commonly used in web development for both client-side and server-side applications.

In this study, JavaScript is used in two major areas:

- Front-end development through React
- Back-end development through Node. js

JavaScript enables the creation of interactive and dynamic user interfaces, allowing real-time updates of lot statuses without requiring page reloads. It also supports asynchronous operations, which are essential for handling API requests efficiently.

The use of a single programming language across both front-end and back-end simplifies development and improves consistency within the system architecture.

3.4.2 Front-End Framework

React. Js

React is an open-source JavaScript library developed by Meta (Facebook) used for building user interfaces, particularly single-page applications. It allows developers to create reusable UI components, making the development process more efficient and organized.

In the proposed system, React is used to build the user interface for the lot monitoring platform. It handles the rendering of:

- Interactive subdivision map
- Dashboard interface
- User login and registration pages
- Forms of updating lot status

One of the key advantages of React is its Virtual DOM, which improves performance by updating only the components that change instead of reloading the entire page. This makes the system more responsive, especially when handling frequent updates in lot status data.

React also supports component-based architecture, allowing the system to be divided into reusable modules. This improves maintainability and scalability.

3.4.3 Styling Framework

Tailwind CSS

Tailwind CSS is a utility-first CSS framework used for designing responsive and modern user interfaces. Unlike traditional CSS frameworks, Tailwind provides pre-defined utility classes that allow developers to style elements directly within the HTML or JSX structure.

In the proposed system, Tailwind CSS is used to:

- Design responsive layouts for different screen sizes
- Style buttons, and navigation components
- Create consistent UI design across the system
- Improve user experience through clean and modern interface design

Tailwind CSS enables faster development because it reduces the need for writing custom CSS. It also ensures design consistency across all system modules.

3.4.4 Back-end Framework

Node. js

Node.js is an open-source JavaScript runtime environment that allows developers to execute JavaScript on the server side. It is built on Chrome's V8 JavaScript engine and is known for its event-driven, non-blocking architecture.

In this system, Node.js is used to handle:

- Server-side logic
- API request processing
- Database communication
- Authentication and authorization

Node.js is particularly suitable for real-time applications like the proposed system because it can handle multiple requests simultaneously without performance degradation.

Express. Js

Express.js is a lightweight web application framework built on top of Node.js. It simplifies the process of building APIs and handling HTTP requests.

In the proposed system, Express.js is used to :

- Create RESTful API endpoints
- Manage routing between frontend and backend
- Handle HTTP request and response
- Implement middleware for authentication and validation

Express.js improves development efficiency by providing a structured and minimal framework for building server-side applications.

3.4.5 Database Management System

MySQL

MySQL is a relational database management system (RDBMS) used to store and manage structured data. It is one of the most widely used databases in web application development due to its reliability, scalability, and performance.

In the proposed system, MySQL is used to store:

- User accounts and credentials
- Lot information and status (available, pending, sold)
- Subdivision or project data
- Activity logs and system records

MySQL uses tables with defined relationships, ensuring data integrity and reducing redundancy. It also supports SQL queries, which allow efficient retrieval and manipulation of data.

The use of MySQL ensures that all lot information is stored in a centralized and secure database that can be accessed in real time.

3.4.6 Email Notification Service

Nodemailer SMTP

Nodemailer is a Node.js module used for sending emails from applications. It supports SMTP (Simple Mail Transfer Protocol), which allows the system to send automated email notifications.

In the proposed system, Nodemailer is used for **:**

- Sending account verification email
- Password recovery notification
- System alerts and updates
- Notification of changes in lot status (if enabled)

Email notifications improve communication between the system and users by providing real-time updates about important actions within the system.

3.4.7 Code Editor and Development Tools

Visual Studio Code (VS Code)

Visual Studio Code is a lightweight but powerful source code editor developed by Microsoft. It is widely used in web development due to its support for multiple programming languages, extensions, and debugging tools.

In this study, VS Code is used for:

- Writing and editing source code
- Debugging application errors
- Managing project files
- Integrating GIT for version control

VS Code enhances productivity by providing features such as syntax highlighting, auto-completion, and integrated terminal support.

3.4.8 Version Control System

Git and GitHub

Git is a distributed version control system used to track changes in source code during development. GitHub is a cloud-based platform used to host Git repositories.

In the proposed system, Git and GitHub are used to:

- Track changes in the source code
- Collaborate among team members
- Maintain version history of the system
- Backup project file securely

Version control is essential in collaborative development environments because it prevents code conflicts and ensures that previous versions of the system can be restored if needed.

3.4 Summary of Tools and Technologies

The combination of React, Node.js, Express.js, MySQL, Tailwind CSS, and Nodemailer provides a complete full-stack solution for the development of the Web-Based Lot Availability Monitoring System.

These technologies were selected because they offer:

- High performance

- Scalability
- Security features
- Flexibility
- Strong community support
- Compatibility with modern web applications

Together, they enable the development of a centralized, secure, and efficient system that addresses the manual limitations currently experienced by Golden Dragon Estate Corporation.

3.5 Data Gathering Procedures

The success of any information system development project depends on the accuracy and completeness of the information gathered during the initial stages of development. For this reason, the researchers conducted a systematic data-gathering process to obtain relevant information regarding the current lot monitoring procedures, operational challenges, user requirements, and organizational needs of Golden Dragon Estate Corporation.

The data gathered served as the primary basis for identifying system requirements, designing system functionalities, and developing features that directly address the company's operational concerns. Multiple data-gathering methods were employed to ensure the reliability and validity of the collected information. These methods include interviews, observation, document analysis, and requirements validation.

Through these techniques, the researchers gained a comprehensive understanding of the existing workflow and were able to identify opportunities for automation and process improvement.

3.5.1 Interview Method

The interview method was one of the primary techniques used to gather information from individuals directly involved in the company's operations. Interviews were conducted with personnel responsible for monitoring lot availability, maintaining subdivision records, and managing sales-related information.

The purpose of the interviews was to obtain first-hand information regarding the current processes used by the organization, identify operational difficulties, and determine the expectations of future system users. The researchers prepared a set of guide questions to ensure that all necessary information was collected while allowing respondents to provide detailed explanations of their experiences and concerns.

During the interviews, participants described how lot availability is currently monitored using printed subdivision layout maps. Lot statuses are manually updated through color-coding techniques, where available lots are marked in green, pending lots in yellow, and sold lots in red. The respondents explained that although the process is familiar to employees, it becomes increasingly difficult to manage as the number of lots and projects grows.

Several concerns were identified during the interview process, including:

- Delays in updating lot status information.
- Difficulty tracking changes across multiple projects.
- Risk of human error during manual updates.
- Possibility of losing or damaging physical records.
- Limited accessibility of lot information.
- Lack of centralized storage for monitoring records.

The respondents also expressed their desire for a system that would allow authorized users to update lot statuses quickly, access information from a centralized platform, and reduce reliance on physical documents.

The information obtained through the interviews significantly contributed to the identification of system requirements and helped ensure that the proposed system aligns with the operational needs of Golden Dragon Estate Corporation.

3.5.2 Observation Method

Observation was conducted to gain a direct understanding of how lot monitoring activities are performed within the organization. Unlike interviews, which rely on verbal explanations, observation allows researchers to witness actual work practices and identify issues that may not be explicitly mentioned by participants.

During the observation process, the researchers examined the procedures used by employees when updating lot information, monitoring lot availability, and retrieving records for customer inquiries. Particular attention was given to the use of subdivision layout maps and the methods employed for updating lot statuses.

The observation revealed several inefficiencies in the current workflow. Employees were required to manually locate specific lots within large printed maps before updating their status. Changes were recorded using colored markers or annotations, which increased the possibility of inconsistencies and accidental errors.

The researchers also observed that retrieving information often required employees to physically search through multiple documents. This process could become time-consuming, particularly when dealing with multiple subdivisions or large numbers of lots.

Additionally, the observation highlighted several risks associated with maintaining physical records, including:

- Wear and tear of printed maps.
- Misplacement of important documents.
- Difficulty maintaining updated versions of records.
- Limited accessibility when documents are not physically available.

These observations reinforced the need for a centralized web-based system capable of providing real-time access to lot information while reducing dependence on manual record-keeping practices.

The findings obtained through observation were used to validate information gathered during interviews and contributed to the overall system analysis process.

3.5.3 Document Analysis

Document analysis was conducted to examine existing records, forms, reports, and other materials currently used by Golden Dragon Estate Corporation in monitoring lot availability.

This method enabled the researchers to understand how information is organized, stored, and utilized within the organization. By reviewing existing documents, the researchers were able to identify the types of data that must be accommodated by the proposed system.

The documents analyzed included:

- Printed subdivision layout maps.
- Lot inventory records.
- Sales monitoring documents.
- Reservation records.
- Customer information forms.
- Internal reports related to lot status monitoring.

Through document analysis, the researchers identified essential data elements required by the system, including:

- Lot number.
- Block number.
- Project or subdivision name.
- Lot area.
- Lot status.
- Reservation information.
- Date of status updates.
- User responsible for modifications.

The analysis also revealed inconsistencies in manual documentation practices, particularly when updates were not immediately reflected across all records. Such inconsistencies increase the likelihood of discrepancies and confusion among personnel.

The findings from document analysis were used to design the database structure and determine the information fields required within the proposed system.

3.5.4 Requirements Gathering and Validation

After collecting information through interviews, observations, and document analysis, the researchers conducted a requirements gathering and validation process. The purpose of this activity was to consolidate the collected information and identify the specific functionalities that the system must provide.

Requirements were categorized into functional and non-functional requirements.

Functional Requirements

Functional requirements describe the specific actions that the system must perform. Based on the collected data, the following functional requirements were identified:

1. User authentication and login functionality.
2. Role-based access control for authorized personnel
3. Lot monitoring and visualization.
4. Interactive lot status updating.
5. Search and filtering capabilities.
6. Dashboard and summary reporting.
7. User account management.
8. Activity logging and record tracking.
9. Email notification services.
10. Database storage and retrieval functionalities.

These features directly address the challenges identified during data gathering and support the operational objectives of the organization.

Non-Functional Requirements

Non-functional requirements describe the quality attributes and performance expectations of the system. The following non-functional requirements were identified:

1. Security – The system must protect sensitive information from unauthorized access.
2. Reliability – The system must consistently perform intended functions without failure.
3. Usability – The interface must be easy to understand and operate.
4. Performance – The system should respond quickly to user requests.
5. Scalability – The system should accommodate future expansion.
6. Maintainability – The system should be easy to update and improve.
7. Availability – Authorized users should be able to access the system whenever needed.

The identified requirements were reviewed and validated against the information obtained from stakeholders to ensure that they accurately reflect the organization's needs.

3.5.5 Importance of Data Gathering in the Study

The data-gathering process played a critical role in the successful development of the proposed system. Through interviews, observations, document analysis, and requirements validation, the researchers acquired a comprehensive understanding of the company's current workflow and operational challenges.

The information obtained during this phase provided the foundation for system planning, analysis, design, development, and evaluation. It ensured that the proposed Web-Based Lot Availability Monitoring System addresses real organizational needs rather than assumptions made by the development team.

Furthermore, the use of multiple data-gathering techniques increased the reliability of the findings by allowing information obtained from one method to be verified through another. This triangulation approach strengthened the overall quality of the study and contributed to the development of a more accurate and effective software solution.

3.6 Testing Procedures

Testing is a critical phase in software development that ensures the developed system functions according to specified requirements and performs reliably under various conditions. The purpose of testing is to identify errors, verify system functionality, evaluate system performance, and ensure that the system meets user expectations.

For the proposed Web-Based Lot Availability Monitoring System, several testing methods were conducted throughout the development process. These testing procedures were designed to evaluate individual system components, the overall integration of modules, and the acceptability of the system from the users' perspective.

The testing activities performed in this study include Unit Testing, System Testing, and User Acceptance Testing (UAT).

3.6.1 Unit Testing

Unit Testing is a software testing method in which individual components or modules of the system are tested independently to ensure that each unit performs its intended function correctly. The primary objective of unit testing is to detect errors at an early stage of development before they affect other parts of the system.

For the proposed system, each major module was tested separately during development. This allowed the researchers to verify that individual functionalities operated correctly before integrating them into the complete system.

The modules subjected to unit testing include:

Authentication Module

The Authentication Module was tested to ensure that users can securely access the system using valid credentials**.**

The following test scenarios were performed:

- Login using valid username and password.
- Login using invalid username.

- Login using incorrect password.
- Logout functionality.
- Session management verification.

Expected results included successful login for valid credentials and appropriate error messages for invalid login attempts.

Lot Monitoring Module

The Lot Monitoring Module was tested to verify that lot information is displayed correctly and that status indicators accurately reflect the information stored in the database.

- Test scenarios included:
- Displaying available lots.
- Displaying pending lots.
- Displaying sold lots.
- Loading lot information from the database.
- Updating visual status indicators.

Expected results included accurate display of lot data and correct color representation for each lot status.

Lot Status Module

This module was tested to ensure that authorized users can update the status of lots successfully.

Test scenarios included:

- Updating a lot from Available to Pending.
- Updating a lot from Pending to Sold.
- Updating a lot from Sold to Available (if permitted).
- Saving status changes.

Expected results included successful database updates and immediate reflection of changes on the user interface.

User Management Module

The User Management Module was tested to verify that administrators can manage user accounts effectively.

- Test scenarios included:
- Adding new users.
- Editing user information.
- Deactivating user accounts.
- Assigning user roles.

Expected results included successful execution of administrative functions without affecting existing records.

Email Notification Module

The Email Notification Module was tested to ensure proper delivery of automated email messages.

Test scenarios included:

- Sending account creation notifications.
- Sending password reset emails.
- Sending system-generated alerts.

Expected results included successful email delivery to the intended recipients.

Dashboard Module

The Dashboard Module was tested to verify that statistical information and summaries are displayed accurately.

- Test scenarios included:
- Displaying total available lots.
- Displaying total pending lots.

- Displaying total sold lots.
- Refreshing dashboard information after updates.

Expected results included real-time updates and accurate statistical summaries.

3.6.2 System Testing

System Testing is performed after all modules have been integrated into a complete system. The purpose of system testing is to evaluate the functionality, performance, and reliability of the entire application as a unified system.

Unlike unit testing, which focuses on individual components, system testing verifies that all modules work together correctly and that data flows properly between different parts of the application.

For the proposed system, system testing focused on the following areas:

Functional Testing

Functional testing was conducted to verify that all system features operate according to specified requirements.

The following functionalities were evaluated:

- User authentication
- Lot monitoring

- Lot status updates
- User management
- Email notifications
- Dashboard reporting

Each feature was tested using various input scenarios to ensure proper functionality.

Integration Testing

Integration testing was performed to verify communication between system components.

The researchers evaluated:

- Front-end and back-end integration
- API communication
- Database connectivity
- Email service integration

This testing ensured that data entered by users is correctly processed by the server and stored in the database.

Performance Testing

Performance testing was conducted to determine the responsiveness and efficiency of the system.

The following factors were evaluated:

- Page loading speed
- Database query performance
- Response time for user actions
- Simultaneous user access

The results indicated that the system performs efficiently under normal operating conditions and responds promptly to user requests.

Security Testing

Security testing was conducted to verify that unauthorized users cannot access restricted functionalities.

Security evaluations included:

- Authentication validation
- Session management testing
- Access control verification
- Password protection mechanisms

The testing confirmed that only authorized users can access and modify system data.

Compatibility Testing

Compatibility testing was performed to ensure that the system functions correctly across different web browsers and devices.

The system was tested using:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Results showed that the application maintained consistent functionality and appearance across supported browsers.

3.6.3 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) is the final stage of testing performed before system deployment. The purpose of UAT is to determine whether the developed system meets the expectations and requirements of the intended users.

For this study, selected personnel from Golden Dragon Estate Corporation were invited to evaluate the system. Participants were asked to perform common tasks and provide feedback regarding the system's functionality, usability, and overall effectiveness.

The evaluation focused on the following areas:

Ease of Use

Users evaluated whether the interface was easy to understand and navigate.

Indicators included:

- Clarity of menus
- Simplicity of navigation
- Ease of performing tasks

Functionality

Users evaluated whether system features performed as expected.

Indicators included:

- Accuracy of lot monitoring
- Correctness of status updates
- Reliability of reports and dashboards

Efficiency

Users evaluated whether the system improved operational efficiency compared to manual processes.

Indicators included:

- Speed of accessing information
- Time required for updating records
- Ease of retrieving lot information

User Satisfaction

Users evaluated their overall satisfaction with the system.

Indicators included:

- Overall experience
- Perceived usefulness
- Willingness to use the system regularly

Feedback obtained during User Acceptance Testing was analyzed and used to implement final improvements before deployment.

The results of UAT served as evidence that the developed system is capable of meeting the operational needs of Golden Dragon Estate Corporation.

3.6.4 Bug Identification and Resolution

Throughout the testing process, issues and software defects were documented and addressed by the development team.

The bug resolution process involved:

1. Identifying the issue.
2. Reproducing the error.
3. Determining the root cause.
4. Implementing corrective measures.
5. Retesting the affected functionality.
6. Verifying successful resolution.

This process ensured that identified problems were resolved before the system was released to users.

3.6.5 Summary of Testing Procedures

Testing played a vital role in ensuring the quality and reliability of the proposed Web-Based Lot Availability Monitoring System. Through Unit Testing, System Testing, and User Acceptance Testing, the researchers were able to verify the correctness of system functionalities, identify and resolve issues, and evaluate the acceptability of the system among intended users.

The successful completion of testing activities provided confidence that the system meets its objectives and is ready for implementation within Golden Dragon Estate Corporation.

3.7 Evaluation Criteria

The developed Web-Based Lot Availability Monitoring System will be evaluated to determine whether it effectively addresses the operational needs of Golden Dragon Estate Corporation. Evaluation is an essential process that measures the quality, effectiveness, and acceptability of the system from the perspective of its intended users.

The evaluation focuses on five major criteria: Functionality, Usability, Reliability, Efficiency, and Security. These criteria were selected because they represent the key characteristics of a successful information system and directly relate to the objectives of the proposed project.

The results of the evaluation will provide valuable feedback regarding the strengths of the system and identify potential areas for improvement.

Functionality

Functionality refers to the degree to which the system performs its intended tasks accurately and completely. It evaluates whether all features and functions operate according to the specified requirements.

For the proposed system, functionality focuses on the ability of the application to perform essential operations such as user authentication, lot monitoring, lot status updating, user management, dashboard reporting, and email notification services.

The evaluation of functionality seeks to determine whether the system successfully addresses the operational challenges currently experienced by Golden Dragon Estate Corporation.

The following indicators may be used:

1. The system correctly displays lot information.
2. The system accurately updates lot statuses.
3. The system generates accurate reports.
4. The system successfully manages user accounts.
5. The system performs all required functions without errors

A high functionality rating indicates that the developed system effectively fulfills its intended purpose.

Usability

Usability refers to the ease with which users can learn, understand, and operate the system. A usable system enables users to perform tasks efficiently with minimal confusion and training.

Since the intended users of the proposed system may have varying levels of technical expertise, usability is a critical factor in determining system acceptance.

The researchers designed the user interface with simplicity and clarity in mind. The use of color-coded lot indicators, intuitive navigation menus, and organized dashboards contributes to a positive user experience.

The following indicators may be used:

1. The system interface is easy to understand.
2. Navigation between pages is simple and intuitive.
3. Information is clearly presented.
4. Users can perform tasks without difficulty.
5. The system is user-friendly.

A high usability rating suggests that users can effectively interact with the system and complete tasks efficiently.

Reliability

Reliability refers to the ability of the system to perform consistently and accurately under normal operating conditions. A reliable system produces dependable results and minimizes unexpected failures or errors.

For the proposed system, reliability is essential because employees depend on accurate lot information when monitoring availability and responding to customer inquiries.

The system must consistently:

- Display accurate lot data.
- Save updates successfully.
- Retrieve information correctly.
- Maintain stable performance during operation.

The following indicators may be used:

1. The system performs consistently during use.
2. Information displayed by the system is accurate.
3. The system successfully saves and retrieves data.
4. The system operates without frequent crashes.
5. The system provides dependable results.

A high reliability rating indicates that users can trust the system to perform its intended functions consistently.

Efficiency

Efficiency refers to the ability of the system to perform tasks using minimal time and resources while maintaining acceptable performance levels.

One of the primary goals of the proposed project is to improve operational efficiency by replacing manual monitoring procedures with an automated digital platform.

The system is expected to reduce the time required to:

- Locate lot information.
- Update lot statuses.
- Generate reports.
- Retrieve records.

The following indicators may be used:

1. The system responds quickly to user actions.
2. Information retrieval is fast and efficient.
3. The system reduces manual workload.
4. The system improves productivity.
5. Tasks can be completed in less time compared to the existing process.

A high efficiency rating demonstrates that the system successfully improves organizational operations.

Security

Security refers to the protection of system resources and information from unauthorized access, modification, disclosure, or destruction.

Because the proposed system contains sensitive business information, appropriate security measures are necessary to protect organizational data.

The system incorporates security mechanisms such as:

- User authentication.
- Password protection.
- Role-based access control.
- Session management.
- Restricted access to administrative functions.

The following indicators may be used:

1. The system protects user accounts from unauthorized access.
2. Sensitive information is secured.
3. Access privileges are properly enforced.
4. User authentication functions correctly.

5. The system maintains data confidentiality and integrity.

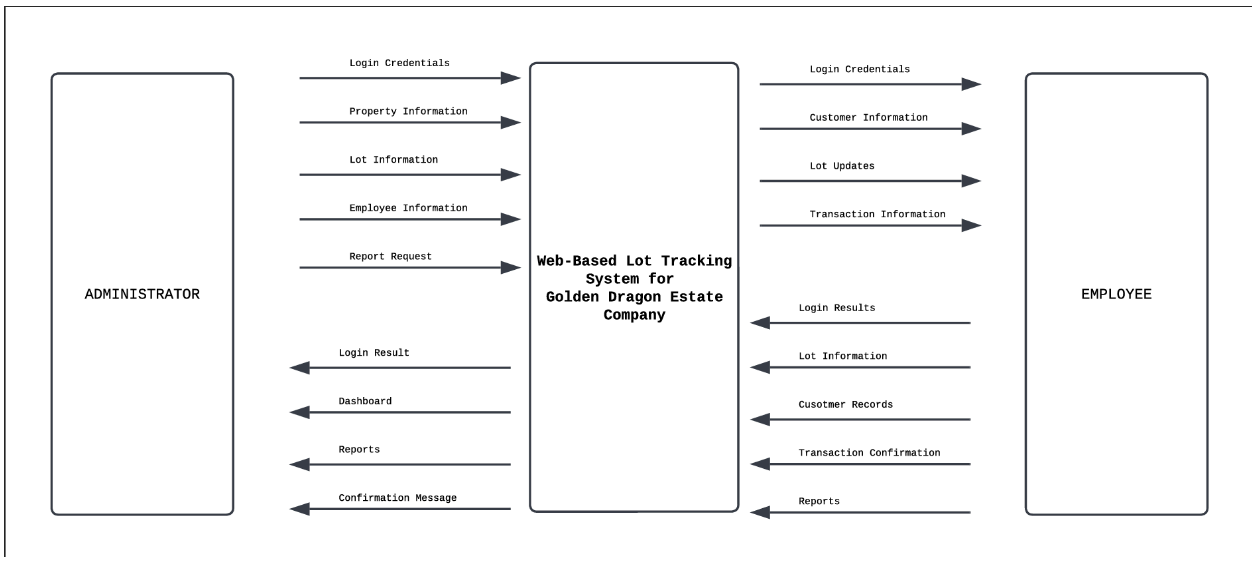
A high security rating indicates that users have confidence in the system's ability to safeguard information.

Summary of Evaluation Criteria

The evaluation criteria provide a structured framework for assessing the quality and effectiveness of the developed system. By examining functionality, usability, reliability, efficiency, and security, the researchers can obtain a comprehensive understanding of system performance.

The evaluation process ensures that the proposed system not only functions correctly but also provides a positive user experience, reliable operation, improved efficiency, and adequate security. The feedback gathered through this process will support continuous improvement and contribute to the overall success of the project.

Data Flow Diagram (DFD) Level 0



The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram, illustrates the overall flow of information between the proposed Web-Based Lot Tracking System and its external entities. It identifies the primary users of the system and the major data exchanged during system operation.

Administrator

- The Administrator serves as the primary user of the system with full access to its features and functions.
- The administrator submits **login credentials** to gain authorized access to the system.

- The administrator manages **property information, lot information, employee information**, and submits **report requests**.
- In return, the system provides the **login result, dashboard, generated reports,** and **confirmation messages** after successful transactions.

Employee

- The Employee interacts with the system according to the permissions assigned by the administrator.
- The employee submits **login credentials**, **customer information**, **lot updates**, and **transaction information** for processing.
- After processing, the system returns the **login results, updated lot information, customer records, transaction confirmations,** and **reports** needed for daily operations.

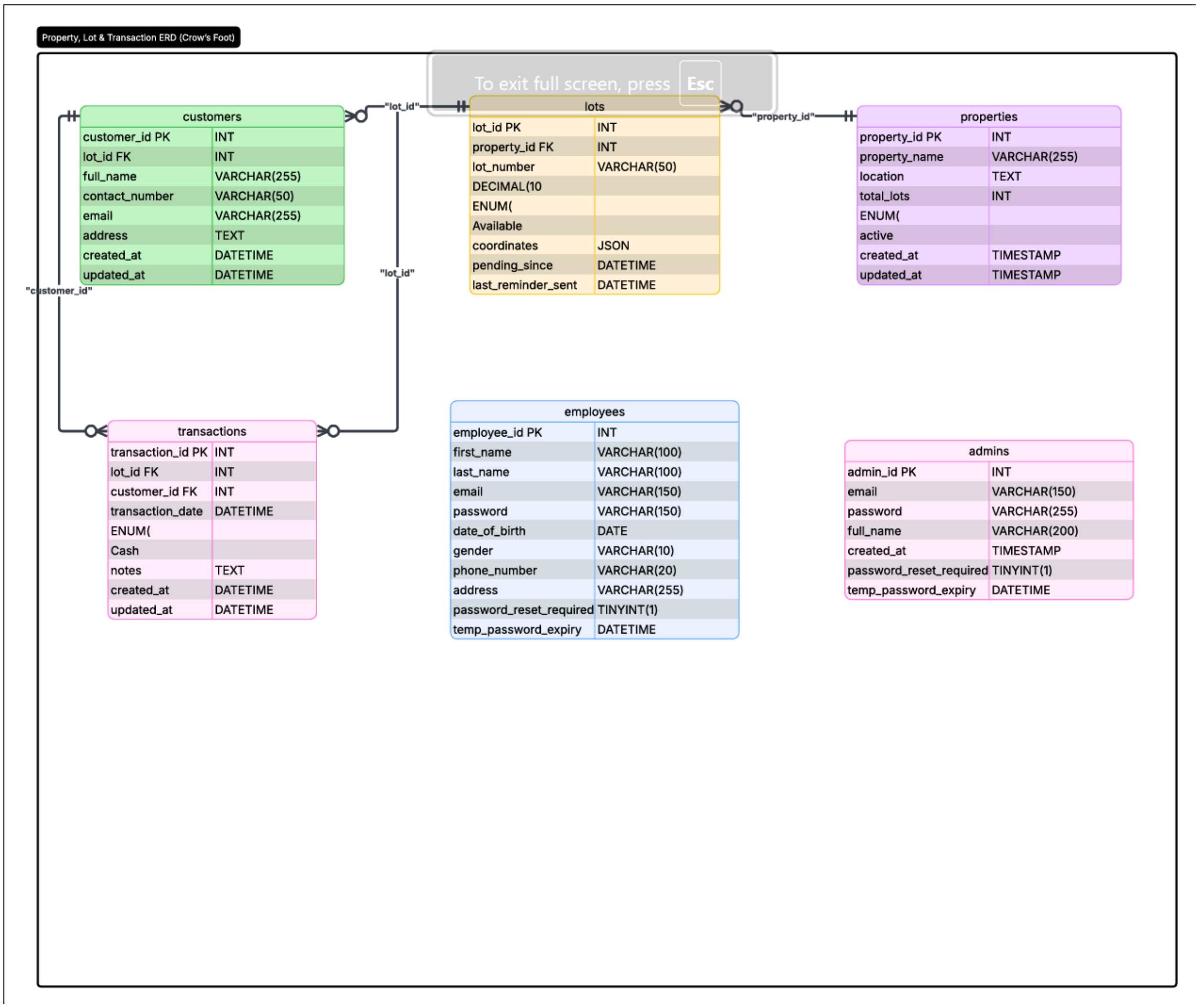
Web-Based Lot Tracking System

- The Web-Based Lot Tracking System acts as the central processing unit that receives, validates, processes, and stores all information submitted by the users.
- It manages the flow of information between the users and the database while ensuring that all lot records remain accurate, updated, and consistent.
- The system also generates reports, processes transactions, and provides real-time updates on lot availability.

Summary

- The Level 0 DFD provides a high-level overview of the proposed system by showing the interaction between the Administrator, Employee, and the Web-Based Lot Tracking System.
- It demonstrates how the system centralizes data processing, improves information management, and supports efficient monitoring of lot availability at Golden Dragon Estate Corporation.

Entity Relationship Diagram (ERD)



The **Entity Relationship Diagram (ERD)** illustrates the database structure of the proposed **Web-Based Lot Tracking System for Golden Dragon Estate Corporation**. It identifies the major entities, their attributes, primary keys, foreign keys, and the relationships among them. The ERD serves as the blueprint for the system's database, ensuring that information is organized, consistent, and efficiently managed.

Customers

- The **Customers** entity stores information about individuals who inquire about, reserve, or purchase residential lots.
- It contains customer details such as full name, contact number, email address, and residential address.
- Each customer is assigned a unique **Customer ID (Primary Key)**.
- The **Lot ID (Foreign Key)** associates the customer with a specific residential lot.

Lots

- The **Lots** entity serves as the core table of the system since it contains all information related to individual residential lots.
- It stores details such as the lot number, current availability status, map coordinates, pending date, and reminder information.
- Each lot is identified by a unique **Lot ID (Primary Key)**.
- The **Property ID (Foreign Key)** connects every lot to its corresponding property or subdivision.

Properties

- The **Properties** entity stores information about the subdivisions or real estate properties managed by the company.
- It contains details such as the property name, location, total number of lots, and active status.
- Each property is identified by a unique **Property ID (Primary Key)**.
- One property can contain multiple residential lots, establishing a **one-to-many relationship** between Properties and Lots.

Transactions

- The **Transactions** entity records all lot-related transactions processed within the system.
- It stores information such as the transaction date, payment method, notes, and references to both the customer and the lot involved in the transaction.
- The table contains **Lot ID** and **Customer ID** as foreign keys, linking every transaction to the corresponding customer and lot.
- This entity provides a complete history of reservations, purchases, and other lot transactions.

Employees

- The **Employees** entity stores the personal and login information of company employees who are authorized to access the system.
- It includes employee details such as name, email address, password, contact information, and address.
- Password reset fields are also included to support account recovery and improve system security.
- Each employee is assigned a unique **Employee ID (Primary Key)** for identification.

Administrators

- The **Administrators** entity stores the login credentials and account information of system administrators.
- Administrators are responsible for managing system users, properties, lots, and other administrative functions.
- The table contains administrator credentials and password recovery information to strengthen system security.
- Each administrator is identified using a unique **Admin ID (Primary Key)**.

Relationships

- One **Property** can have **many Lots**, while each lot belongs to only one property.
- One **Lot** can be associated with a customer during reservation or purchase.
- One **Customer** can have one or more recorded transactions depending on the business process.
- Each **Transaction** is linked to a specific customer and residential lot to maintain accurate transaction records.
- **Employees** and **Administrators** manage the system through authenticated accounts, ensuring secure access to system functions.

Summary

- The Entity Relationship Diagram provides the logical database design of the proposed system by illustrating how data is organized and connected.
- It ensures data integrity through the use of primary and foreign keys while minimizing redundancy by maintaining well-defined relationships among entities.
- The database structure supports efficient storage, retrieval, and management of lot information, customer records, property details, and transaction history, enabling the Web-Based Lot Tracking System to provide accurate and reliable information for Golden Dragon Estate Corporation.

The **Use Case Diagram** illustrates the interactions between the users of the proposed **Web-Based Lot Availability Monitoring System for Golden Dragon Estate Corporation** and the different functions available within the system. It identifies the two primary actors, namely the **Administrator** and the **Employee**, and defines the system functionalities that each user can access based on their assigned roles and permissions. The system implements **Role-Based Access Control (RBAC)** to ensure that users can only perform actions authorized for their respective roles.

Administrator

- The **Administrator** has full access to all system functions and is responsible for managing the overall operation of the application.
- The administrator must **log in** before accessing any protected system features.
- The administrator can **manage employees** by creating, updating, or removing employee accounts.
- The administrator can **manage properties** by adding, editing, or maintaining property information.
- The administrator can **manage lots** by updating the availability status, lot details, and other related information.
- The administrator can **view customer records** for monitoring customer information.
- The administrator can **view transaction records** to monitor reservations, purchases, and other lot-related activities.

- The administrator can **view the dashboard** to monitor the overall status of properties and lot availability.
- The administrator can **generate reports** to obtain summaries of lot status, customer information, and transactions.
- After completing system operations, the administrator can **log out** to securely end the session.

Employee

- The **Employee** accesses the system using authorized login credentials.
- Employees can **manage lots** by updating the status of lots assigned to them during daily operations.
- Employees can **manage customer information** by recording and updating customer details.
- Employees can **record transactions** involving lot reservations and sales.
- Employees can **view the dashboard** to monitor current lot availability and assigned activities.
- Employees can **generate reports** relevant to their daily operations.
- Employees can **log out** after completing their assigned tasks.

Common System Functions

- **Login** authenticates the identity of both administrators and employees before granting access to the system.
- **View Dashboard** provides users with a summary of important information, including lot availability, customer records, and recent transactions.
- **Generate Reports** allows users to produce reports for monitoring and decision-making purposes based on their access level.
- **Logout** securely terminates the user's active session to protect system data from unauthorized access.

Role-Based Access Control (RBAC)

- The system implements **Role-Based Access Control (RBAC)** to regulate user permissions.
- Administrators are granted complete access to all system modules, including employee management, property management, lot management, customer records, transaction monitoring, and report generation.
- Employees are provided access only to the modules necessary for their daily responsibilities, such as managing customers, updating lot information, recording transactions, and generating operational reports.

- This access control mechanism enhances system security by preventing unauthorized users from accessing sensitive administrative functions.

Summary

- The Use Case Diagram illustrates the functional requirements of the proposed system by identifying the interactions between users and the available system features.
- It clearly defines the responsibilities of both administrators and employees while demonstrating the implementation of Role-Based Access Control (RBAC) to ensure secure and organized system operations.
- Through these interactions, the Web-Based Lot Availability Monitoring System enables efficient management of properties, residential lots, customers, and transactions, thereby improving the overall operational efficiency of Golden Dragon Estate Corporation.

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